उत्तराखण्ड राज्य
DRONE APPLICATION & RESEARCH CENTRE
नभः अनिमिषत्
DRONATHON @ DRONANAGRI 2020
INDIA DRONE
FESTIVAL 2.0
FEB 19-20,2020,ITDA,IT PARK
DEHRADUN,UTTARAKHAND

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1. MESSAGE OF HONOURABLE CHIEF MINISTER

Honourable Chief Minister Shri Trivendra Singh Rawat has inaugurated the India Drone Festival 2.0. Giving his opening address, the Chief Minister congratulated and gave best wishes to everyone on successfully organising 2nd edition of Drone festival – Dronathon@ Dronagiri 2020.

Speaking on the occasion, he said that, with the help of NTRO, the state of Uttarakhand has made considerable progress in drone technology and application space. He also said that, he is very confident that this Drone festival will help students understand drone technology, its utility and importance. This festival will also inspire many new talents in exploring this technology and may encourage students to pursue this & related area as their profession.

The Chief Minister has mentioned that Drone technology has many utilities in states like Uttarakhand, which has very difficult geographical terrain and prone to natural calamities. Drones can play very important role during natural disasters and in emergency health services. The use of this technology can lend a new direction to life, he said.

Referring to the youth as the future of the nation The Chief Minister stressed on the participation of youth in such events. Stating that the nation's first drone application training and research centre was established in Dehradun, he said that by using such modern technologies the youth can play a vital role in the development of the state and nation.

The Chief Minister has also released the GIS based drone mapper software 'DARC Mapper' on the occasion. Prepared by the drone application centre, this software will enable photo capture, making of three dimensional models and data analysis. He also launched 'MyGov Uttarakhand' on this occasion.

2. MESSAGE OF HONOURABLE GOVERNOR

The Honourable governor Baby Rani Maurya has congratulated and conveyed her best wishes on organising Drone Festival.

Addressing the gathering, she said that given the difficult terrain, drone technology can play a very significant role in disaster management and relief. She also said that drone technology has many applications, and we should explore and take maximum benefit of this technology. However, every technology has both positive and negative sides, and we should be worried about the negative aspects as well.

Addressing students, she said that we should focus on positive aspects of this technology and should apply the same in right direction. She said that this event will increase the awareness in Drone technology space.

The Governor recalled that the nation's first drone application centre had been established in Uttarakhand. Drone technology should be used for the development and benefit of remote mountainous areas of the state to bring about qualitative improvement in the lives of the local residents.

She said that she is confident that this event must have provided the platform for ideas exchange in this space.

In the context of Uttarakhand, good use of this technology can be made for works related to disaster management, education, medical and health. The help of drone mounted cameras can be taken in situations wherein swift assistance is required in remote areas of the state.

Referring to the projected growth of the industry, she said that this sector also provides good employment opportunities for the youth. She said that the application of drone technology is being seen in photography, which is also generating employment.

She said that Importance of drone technology is also increasing in defence & strategy and experts should consider that this technology should be used only for positive purposes and not be misused.

She concluded her speech by highlighting that such events will create awareness and promote research & development in Drone space and congratulated everyone for making this event a great success.

3. MESSAGE OF INFORMATION TECHNOLOGY SECRETARY

IT Secretary Shri. R.K. Sudhanshu thanked everyone for their participation in Drone festival – Dronathon@ Dronagiri 2020, specially the students, companies and media personal.

Speaking on the occasion, the IT Secretary briefly explained about the 2nd edition of Drone festival – Dronathon@ Dronagiri 2020. He mentioned that during this two day Drone festival, more than 160 contestants participated in the various drone competitions. Many national and global organizations have also made their presence felt by spreading awareness about drone technology and its applications through their presentation, stalls, and live application demonstrations.

The IT Secretary has mentioned that Uttarakhand is one of the pioneers in the field of drone technology and is the first state to establish a drone research Centre. DARC has trained many people, including many students & professionals, not only from Uttarakhand but also from many states of the country, in drone technology and its applications. He emphasized that Uttarakhand has made tremendous progress in the area of Drone Technology.

The IT Secretary also mentioned about the launching of Mygov – Uttarakhand application, which will promote and facilitate citizen's participation in policy formulation. ITDA launched a Drone mapping application during the event as well. One of the key themes of the festival was counter drone technology, which was successfully demonstrated in the event. The event also saw an enthusiastic participation from many school children in the event.

The IT Secretary concluded his message by emphasizing on the human resource development, for which Uttarakhand is known for. He reinstated government's commitment and support in the area of Drone technology, specifically focusing on continuous improvement in the area of production, application research and training.

4. DIRECTOR'S MESSAGE

The ITDA - State's Nodal Agency for IT - is making tremendous efforts towards inclusion and deployment of ICT in key areas of governance and in turn improve the public service delivery.

The ITDA has been the front runner on adopting new technologies. One of the key focus for the ITDA has been Drone - unmanned aerial vehicle. The ITDA is the pioneer in this space and has been the first government agency to established a centre – DARC - to specifically focus in UAV space. DARC has made an incredible progress in last few years and provides a platform where experts can share knowledge and skills on design, utilization and applications of Unmanned Aerial Systems (UAS).

Considering the rapid technological advances in UAV field, and its larger impact on normal citizen's life & society as a whole, it is imperative to bring all the key stakeholders on one platform. Keeping that in mind ITDA & DARC has organised the 2nd edition of "India Drone Festival 2.0 — Dronathon @ Dronagiri ", and the theme of this edition of event was **"Counter-drone Technology" -** issues related to counter drone technologies and creation of intelligent solutions in future.

One of the key objective of this event was bringing the drone technology under one roof to enable rapid innovation, faster growth and global outreach. It would help to create awareness and build a safer, scalable, sustainable flying environment in India.

This event has been a great success this year as well, and we saw enthusiastic participations from more than 20 companies and many drone enthusiast. We will continue to grow in this new technology space and wish to make this event a permanent feature, creating a platform for fruitful interaction between companies, entrepreneurs, and various wings of government.

5. MESSAGE OF BASE DIRECTOR NTRO

Grp Captain Rajat Dwivedi (Retd.) - Base Director – NTRO started his message by congratulating everyone on successful completion of Drone festival – Dronathon@ Dronagiri 2020.

He mentioned that Drone Technology is one of the most important technology innovations and has a big role to play in this technological age. Keeping that in mind, Uttarakhand Government through ITDA & NTRO have collaborated and experimented with establishing DARC – Drone Application and Research Centre on 9th July 2018 to promote and nurture research and development in the field of drone technology.

The journey started two years back, and during the period DARC has come a long way and achieved many successes during this fruitful journey.

He mentioned that he is proud to be part of this journey and feel immense pride in successfully organising this event 2nd time in collaboration with ITDA. He also proudly announced that DARC has given training to more than 250 personnel.

He concluded his message by thanking Honourable Governor, special dignitaries, and all members of ITDA & DARC.

6. FOREWORD

The Second edition of "India Drone Festival organised by Information Development agency (ITDA) of the state government of Uttarakhand, serves as a commendable initiative towards assisting in developing the roadmap of enhancement of technical capability in the field of drone application for trainees & researchers that really matters for our country. On behalf of BDO India LLP, I am glad to extend our support to this event.

Dronathon 2.0 not only highlights the dire need of the Drone Industry in the country but also creates a whole platform for pooling in different ideas and suggestions of young talents from different walks of life. As a consultancy firm, BDO India LLP is delighted to be associated with this event as its knowledge partners. Dronathon is an initiative to build such a network; it started as a mere discussion and ensures to make a substantial contribution in bringing the suggestions of these discussions to practice.

It is truly an honour to partner with and contribute to such an agile and informative event.

Regards,

Ashish Gangrade
Partner
Business Advisory Services
BDO India LLP

Dronathon 2.0 is another promising step by ITDA in collaboration with DARC. Dronathon 2.0 acted as a platform wherein experts from Industry came forward to showcase existing technology, its usage and the challenges they are facing for providing services to nation. This platform was beautifully leveraged by young kids to learn with fun with events like Drone Racing Championship.

Drone which has been used previously by Armed Forces, now finding its place in every segment of life. It is a now handy tool for law enforcement agencies, government bodies, academic institutes and providing services at the time of need. In the current COVID 19 situation, drone has been used extensively by law enforcement agencies to gather information which was further utilized to maintain law and order.

DARC has made incredible progress over a period of time in providing training, sensitization of government agencies, anti-drone and image analysis. The time has come where DARC should play key role and support Government of Uttarakhand as a new hub for Drone Technology.

We would like to thank ITDA, for providing us an opportunity to be associated as Knowledge Partner for Dronathon 2.0.

Regards,

Anurag Srivastava
Associate Partner
Business Advisory Services
BDO India LLP

7. PREAMBLE

Vision: "To use IT as a vehicle for economic development of State of Uttarakhand with all round inclusive growth to create a vibrant society with high quality of life."

Changing technology landscape, and easy access to technology have created a robust ecosystem, facilitating the direct communication between Government and Citizen. These advancements have helped governments formulate, implement, deliver and monitor projects effectively and efficiently, benefitting citizens. Uttarakhand government has taken many initiatives, specifically focusing on Information and Communication Technology, to achieve its vision of creating a vibrant society with high quality of life.

To achieve its vision, Government of Uttarakhand intends to harness the full power of Information and Communication Technology (ICT). The key objectives government wants to achieve through ICT interventions -

- bring in accelerated social and economic development
- ensure transparency in the Government decisions
- accelerate the IT adoption amongst various user segments

The end objective, the government looking forward to, is to achieve an **ideal e-society model** through an efficient, service oriented, cost-effective, information networked, eco-conscious, and year-on-year growth approach.

The Uttarakhand State policy – enable **Uttarakhand to emerge as the Global/National Hub and destination for IT, ITES by 2025.**

Twin goals

- Bring the full power of IT within the reach of the state
- Harnessing the capability and human resources of the state

The ITDA - State's Nodal Agency for IT - is continuously working on successful inclusion and deployment of ICT in all key areas of the economy in the State. One of the key focus area for IT department is to focus on e-Governance for improving public service delivery. It enables the citizen to avail various services at their doorsteps, which is really the need of time. The Government of Uttarakhand (GoU) has planned several projects and tirelessly working in the direction to reach to the citizens.

The department is striving to provide

- A platform to share/ provide information,
- Backend computerization to respond/process the queries/ requests, and
- A delivery mechanism.

Some of the key projects which ITDA is focusing on are Common Service centre, State Data Centre – Phase II, e-District, UKSWAN, National Information Infrastructure- Haridwar, Bharatnet Phase-II, CM Dashboard, CM Helpline (1905), E-Office Implementation, STEP-UK and Drone application and research.

The Uttarakhand Government is also working towards its ambitious Vision 2030, To successfully achieve its objectives, the government has specifically provisioned and created a roadmap of ICT infrastructure in state. Here are the key components of ICT roadmap -

- Mobile connectivity across the state
- Internet connection to all GPs
- Permission to private ISPs to provide connectivity on leased line
- Disaster warning through SMS
- Automated grievance redressal mechanism for all departments

- Improving Digital Literacy in state
- Infrastructure and Industries support to boost ICT&E industry in state
- Provisioning of G2C services through internet and mobile applications
- State departments to undergo GPRs and automation of offices
- Inter-departmental G2G services to be done electronically
- Cashless payment for all G2B and G2C services

The ITDA has been a front runner on exploring and adopting new technology areas. One of the key focus for the ITDA has been Drone - unmanned aerial vehicle. Uttarakhand government has been the pioneers in this area and among the first states to established DARC to specifically focus in UAV space. DARC's vision is to be a place where experts in drone technology and industrial applications of drones can share their knowledge and skills with the government departments and others.

The DARC-Drone Application Research Centre is now the voice and need for UAV Training, Research, Skill Building, Disaster management, Forestry, Mining, Citizen Specific services for Government of Uttarakhand.

Considering the rapid technological advances in UAV field, and its larger impact on normal citizen's life & society as a whole, it is imperative to bring all the key stakeholders on one platform. Keeping that in mind ITDA & DARC organised the 2nd edition of "India Drone Festival – Dronathon @ Dronagiri 2.0". Theme of this edition of event was **"Anti-drone Technology".**

This two days long event was inaugurated by honourable Chief Minister Shri. Trivendra Singh Rawat. More than 20 companies, various departments and students participated in the event. The event saw exchange of ideas, and discussion around challenges faced by the drone industry and possible solution and policy initiative that the government can help with. Several drone-related competitions, focusing on speed, control, and other skills of the participants while operating drones were organised. The honourable governor Baby Rani Maurya presided over the conclusion of the two-day India Drone Festival 2.0.

8. ITDA – INFORMATION TECHNOLOGY DEVELOPMENT AGENCY- GOVT OF UTTARAKHAND

The ITDA - State's Nodal Agency for IT - is continuously working on successful inclusion and deployment of ICT in all key areas of the economy in the State. One of the key focus area for IT department is to focus on e-Governance for improving public service delivery. It enables the citizen to avail various services at their doorsteps, which is really the need of time. The Government of Uttarakhand (GoU) has planned several projects and tirelessly working in the direction to reach to the citizens.

The department is striving to provide -

- A platform to share/ provide information,
- Backend computerization to respond/process the queries/ requests, and
- A delivery mechanism.

Key objectives of ITDA -

- Encourage the use of IT in the Government
- Upgrade the quality of life of citizens by facilitating easy access to consumer applications
- Encourage private sector initiatives for the development of world class IT infrastructure
- Upgrade and develop manpower skills required for the IT industry and to accelerate the use of IT in schools, colleges and other educational institutions
- To use IT as GDP growth driver

Some of the key projects which ITDA is focusing on are –

Devbhoomi Jansewa Kendra (CSC) - e-wallet services, Aadhar registration, & banking services	State Data Centre - Phase II	e-District project – Janadhar Kendras
UKSWAN	Aadhar	Video Conferencing Facility up to Tehsil/Block Level
National Information Infrastructure - Haridwar	Bharatnet Phase-II	Secretariat LAN Upgradation
CM Dashboard	CM Helpline (1905)	E-Office Implementation
e- Gate Pass	Digi Locker	STEP-UK [Skill Trainied For Enhancing Employability & Productivity] CALC 2.0

9. DARC - DRONE APPLICATION AND RESEARCH CENTRE

Drone Application and Research Centre (DARC) is a joint initiative of Information Technology Development Agency (ITDA), Govt. of Uttarakhand and National Technical Research Organization (NTRO). The center has been raised with the intent to promote and nurture research and development in the field of drone Technology.

DARC provides a platform where experts in drone technology can share their knowledge and skill on design, utilization, and applications of Unmanned Aerial Systems (UAS). DARC was inaugurated by Honourable Chief Minister of Uttarakhand, Shri Trivendra Singh Rawat on July 9, 2018.

Its aim is to help government officials, on field officers, individuals and organizations understand the revolutionary benefits of UAS technology and gain the skills to implement drone (UAV) technology safely and effectively in their workplaces and wider enterprises.

DARC was established with the following objectives -

- Establishment of State of Art Centre for Drone Application & Research.
- Establishment of High-Tech Training facility for training of Drone pilots.
- Set up technical facility to develop the capability of Drone Application in the fields of Forest Survey, Disaster Relief operation by police.

Development of this facility lead to the enhancement of technical capability, provided solutions to technical challenges and training for future leaders in the UAV industry.

DARC will render help/assist the users to make informed decisions about drone challenges and thus lead to the cost-effective management of drone integration and adoption.

10. SECTOR OVERVIEW - DRONE INDUSTRY

[1]Drones have a long history and the earliest recorded use of a Drone was in warfare. Technology is changing rapidly and making its access to everyone at low costs. It's very common to see drones in the hands of kids, students and normal person on the street nowadays, and they are potentially as disruptive as the Internet. Usage of drones will fundamentally change the way, we relate to each other across space, across territory and across the city. E-commerce and retail industry are already working on using drones in delivery of product to customers in cost effective and timely manner. Over the years usage of drones has grown in industries like agriculture, construction, Energy, Entertainment and Oil and Gas and in other applications like inspection, monitoring, surveying and mapping and it is fuelling the demand for drones across the globe. Below chart represent the distribution of drone usage across industry.

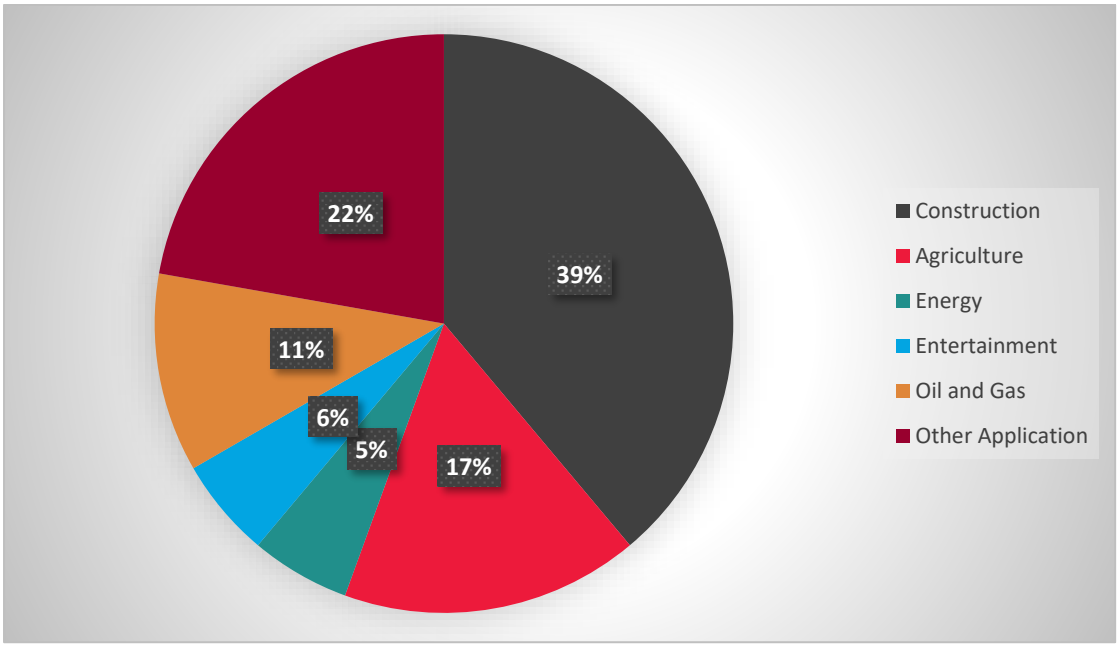


Figure 1-Drone Usage Distribution - Industry wise

Businesses are deploying drone in unprecedented numbers, as usage for unmanned aircraft move beyond photography and venturing into mapping, gathering data and even tracking criminal suspects. In the telecom industry, where workers climb up treacherous poles for inspections, drones are now being deployed to assess them from the air instead. At oil and gas companies the UAVs are being used to conduct monthly pipeline survey. Commercial drone companies have long operated in countries such as China, Japan, Rwanda, and Switzerland—for the delivery of medical and other supplies as well as for agricultural usage.

The drone market in Asia-Pacific is anticipated to witness the highest growth. The recent changes in drone regulations in several countries across the region are expected to support the growth of the market. China became the global hub for drone manufacturing in the past six years. About 70% of the global civilian drone market is supported by China.

Emerging Business can grow beyond expectations if they have a proper transportation infrastructure to deliver their services. Using Drone as a delivery management system will not only multiply their numbers but will also cater to the needs of millions of people at a faster rate. Here are some of the examples of Business Giants who have already started experimenting with the Drone technology: -

- DHL began testing unmanned aerial vehicle (UAV) delivery services in 2014, when it began delivering medical supplies to Just, an island in the North Sea off Germany.
- UPS has successfully tested a drone that launches from the top of a delivery truck.

[1] *Jungleworks.com - Future of Drone Industry*

- Pizza by drone became a reality in November 2016, when Dominos delivered pizza to a customer in Whangaparaoa, New Zealand.
- Amazon made commercial drone delivery to a remote UK farm, and its first US test ran in 2017.
- In July 2018, China's biggest online retailer, JD.com's drone took off from a playground in the city of Xi'an to deliver one of the orders in a football-sized box to a village in the mountains to the south.
- Google parent company Alphabet's drone delivery tests are taking place in a semi-rural residential development In New South Wales, Australia.

Not only these,

- One of India's biggest food-tech firms "Zomato" is all set to form its drone-based delivery network in the country.
- Facebook is researching the possibility of delivering wireless internet to developing areas through drone technology.
- In recent years two innovative ideas came from Facebook and Alphabet, where both the companies have similar approach of delivery internet connectivity to remote location using drones and high-altitude balloons
- Drones are being used by power industry to inspect the solar panels, where they can fly over and check the heat spots on them and work out which solar panels are dead or alive
- Drones are being used by architects to aerial scan and get 3D model with a texture map of the location which is not possible by any other means
- Coal mining companies in India have started aerial surveys of coal blocks for assessment of greenery restoration post excavation from mines
- High-tech drones allow farmers, and the drone pilots that operate them, to increase efficiency in certain aspects of the farming process. From crop monitoring to planting, livestock management, crop spraying, irrigation mapping, and more.
- In the pandemics and outbreaks, drone with an automated sprayer can be used to sanitise large areas including roads, parks, and footpaths, to prevent the spread of coronavirus.
- The Wing drone delivery system developed by Google is being used to deliver toilet paper and medicine to residents in lockdown in Virginia, USA

Drone technology is a phenomenal innovation that continues to have far-reaching effects across today's society, transforming our lives and the way we do business. While these possibilities may be years away, drone technology has already had an impact on how businesses operate. What connects many of these efforts is a need to get goods to consumers while enabling productivity and profit gains. These drone delivery systems will not only empower emerging economies but also will enable them to leapfrog conventional infrastructure investments.

10.1. Drone Industry in India

[2]The drone space in India is catching up with that in other nations and gaining considerable momentum. Government of India, Ministry of Civil Aviation issued Regulations effective from 1st December 2018 on Operations of Remotely Piloted Aircraft System (RPAS) to be enabled through Digital Sky Platform.

The Ministry of Civil Aviation has been working for several years to establish a world leading drone ecosystem in India. To that end, it was necessary to develop global standard drone regulations that would permit appropriate safeguards to commercial application of various drone technologies.

The preparation of these drone regulations through a Civil Aviation Requirement (CAR) - a novel system of software-based self-enforcement to minimize deviations from the CAR - has taken multiple years because:

- Drone technologies have been evolving very rapidly
- Many countries are still experimenting with their drone regulations and no ICAO stands have been developed; and

[2] *pib.gov.in- Regulations for Drones*

- India's security environment necessitates extra precautions

Instead of simply digitizing a paper-based process for registering and operating drones, India has formulated an all-digital process. The Digital Sky Platform is the first-of-its-kind national unmanned traffic management (UTM) platform that implements "no permission, no take-off" (NPNT). Users will be required to do a one-time registration of their drones, pilots, and owners. For every flight (exempted for the nano category), users will be required to ask for permission to fly on a mobile app and an automated process permits or denies the request instantly. To prevent unauthorized flights and to ensure public safety, any drone without a digital permit to fly will simply not be able to take-off. The UTM operates as a traffic regulator in the drone airspace and coordinates closely with the defence and civilian air traffic controllers (ATCs) to ensure that drones remain on the approved flight paths.

[3]Adoption of UAS is increasing rapidly in India and the value of industry and market is projected to be around US$ 885.7 million, while the global market size will touch US$ 21.47 billion by 20212.

10.2. Drone Categories in India

[4]In India, department has primarily categorised Drones in 5 categories, that are listed below. Registration is required for all but the Nano category.

- Nano: Less than or equal to 250 grams (.55 pounds)
- Micro: From 250 grams (.55 pounds) to 2kg (4.4 pounds)
- Small: From 2kg (4.4 pounds) to 25kg (55 pounds)
- Medium: From 25kg (55 pounds) to 150kg (330 pounds)
- Large: Greater than 150kg (33 pounds)

India's Directorate General of Civil Aviation (DGCA) has also initiated programme to test safe integration of beyond visual line of sight (BVLOS) drone flights.

Google-backed hyperlocal delivery start-up Dunzo and Bengaluru-based drone maker Throttle Aerospace Systems have been finally approved by the DGCA based on their experimental programme called beyond visual line of sight (BVLOS) drone operations. Throttle Aerospace Systems has tied up with SpiceJet's dedicated air cargo service SpiceXpress that will provide the company with drones and assist in the initial testing phase.

10.3. [5]Recent developments regarding drones in India

- A specialized force constituted "for the purpose of specialist response to a threatening disaster situation or disaster" has been using drones for locating victims of natural disaster.
- Indian Railways is using UAS for inspection and 3D mapping to bring to life its vision of a dedicated freight corridor with a network of 3,360 km. The entire corridor will be mapped using UAS technology.
- In November 2019, Coal India used drones to check illegal mining and pilferage, a system which it had already tested in a pilot project.
- An autonomous agency of the Government of India, responsible for management of a network of National Highways has employed the use of drones for 3D digital mapping for Detailed Project Report (DPR) for road widening of the Raebareli – Allahabad Highway.

Though both the industry and the market in India are at a very nascent stage, there is an immense potential for growth in both, UAV market in India is projected to grow at a CAGR of 18% by 2023.

[3] *investindia.gov.in- The growing market for drone technologies in India*
[4] *Digitalsky.dgca.gov.in/- Drone Categories in India*
[5] *investindia.gov.in- The growing market for drone technologies in India*

10.4. Predictions

India Commercial Drones Market is projected to grow at a double digital CAGR. Commercial Drones are significantly gaining adoption across several areas owing to growing technological advancements and rising demand for drones in media and entertainment applications. India is among top countries across the globe in terms of Drones imports, India accounts for over 20% of the global Drones imports. Moreover, the government of India is pushing the "Make in India" initiative to increase domestic manufacturing of these drones in the coming years. Based on type, the market has been segmented into Fixed Wing, High-Altitude Long Endurance (HALE), Unmanned Combat Aerial Vehicle (UCAV), Medium-Altitude Long-Endurance Unmanned Aerial Vehicle (MALE) and Others. In 2018, MALE segment dominated the market and is anticipated to maintain its dominance over the next five years as well. Commercial drones have several end user applications like Agriculture & Forestry, Media & Entertainment, Mapping & Monitoring and Others which will drive the overall commercial drone market during the forecast period owing to increasing demand from these sectors. Additionally, the introduction of recent government policies has further augmented the demand for commercial drones in the country.

[6]The beyond line of sight (BLOS) segment of the UAV market is projected to grow at the highest CAGR from 2019 to 2025

The UAV market has been segmented on the basis of UAV type, vertical, class, industry, system, mode of operation, range, point of sale, MTOW, and region. Based on range, the UAV market has been classified into visual line of sight (VLOS), extended visual line of sight (EVLOS), and beyond light of sight (BLOS). The beyond line of sight (BLOS) segment of the UAV market is projected to grow at the highest CAGR from 2019 to 2025. This growth can be attributed to the increasing use of these UAVs in commercial and consumer applications.

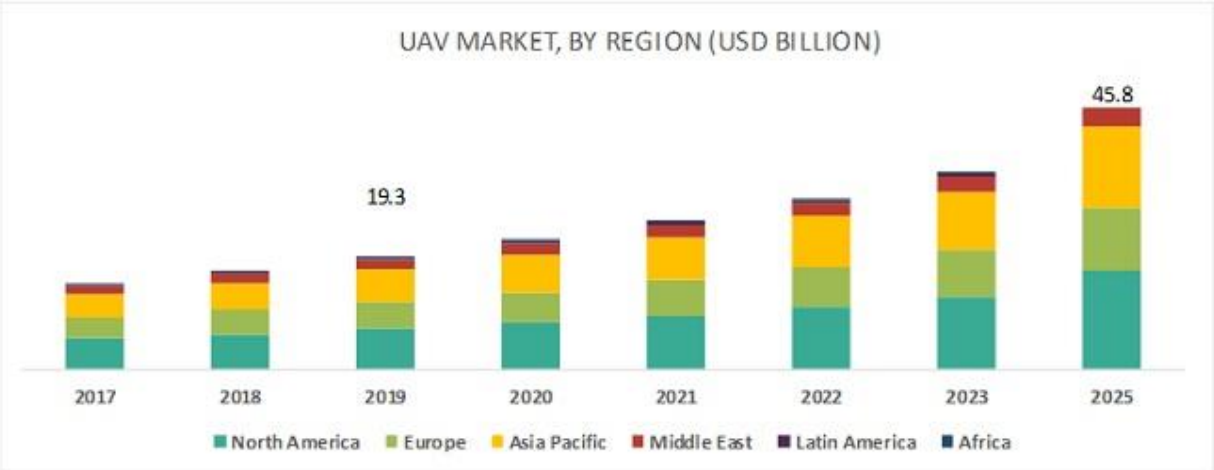


Figure 2 - UAV Market, By Region

The commercial segment of the UAV market is projected to grow at the highest CAGR from 2019 to 2025

Based on application, the UAV market has been segmented into military, commercial, government & law enforcement, and consumer. The commercial segment of the UAV market is projected to grow at the highest CAGR from 2019 to 2025. This growth can be attributed to the increased adoption of UAVs for various commercial applications, such as product delivery, monitoring, surveying, mapping, and remote sensing.

The small UAVs segment is expected to grow at the highest CAGR during the forecast period

Based on class, the UAV market has been segmented into small UAVs, strategic & tactical UAVs, and special-purpose UAVs. The small UAVs segment is expected to grow at the highest CAGR during the forecast period, owing to the demand for these UAVs from the defence sector for surveillance and reconnaissance applications. The demand for small UAVs from the commercial sector is also increasing, as these UAVs are used in precision agriculture, logistics & transportation, wildlife survey, search & rescue operations, firefighting, law enforcement, photography, and disaster management.

[6] *marketsandmarkets.com- (UAV) Market by Vertica*

11. INDIA DRONE FESTIVAL 2.0 -DRONATHON @ DRONANAGIRI

Technological advances have made Drone technology accessible to everyone, and in-turn has opened avenues for using the drone in many uncharted areas. Usage of Drone has special significance in states like Uttarakhand which has difficult geographical terrain and prone to natural calamity.

Government of Uttarakhand has established **DARC** to promote and nurture research and development in the field of drone technology and provide a platform where experts can share knowledge and skills on design, utilization and applications of Unmanned Aerial Systems (UAS). The DARC-Drone Application Research Centre is now the voice and need for UAV Training, Research, Skill Building, Disaster management, Forestry, Mining, Citizen Specific services for Government of Uttarakhand.

Considering the rapid technological advances in UAV field, and its larger impact on normal citizen's life & society as a whole, it is imperative to bring all the key stakeholders on one platform. Keeping that in mind ITDA & DARC organised the 2nd edition of "India Drone Festival 2.0 – Dronathon @ Dronanagiri ". Theme of this edition of event was **"Counter-drone Technology".**

This two days long event was inaugurated by Honourable Chief Minister Shri. Trivendra Singh Rawat. More than 20 companies, various departments and drone enthusiasts participated in the event. The event saw exchange of ideas, and discussion around challenges faced by the drone industry and possible solution and policy initiative that the government can help with. Several drone-related competitions, focusing on speed, control, and other skills of the participants while operating drones were organised. The honourable governor Baby Rani Maurya presided over the conclusion of the two-day India Drone Festival 2.0.

Please refer the section below which highlights the key speaking sessions & competitions of the event.

India Drone Festival 2.0 – Dronathon @ Dronanagari 2020	
DAY-1 - 19th February 2020	**DAY-2 - 20th Feb 2020**
Speaker Panel Sessions	
▶ Mr. Hillol Biswas (DGCA) ▶ Mr. Sunny Sharma (IIO Technologies) ▶ Mr. Luciano Jardim (Milkor India) ▶ Mr. Aditya Thapliyal (ideaForge Technology) ▶ Mr. Ganesh Naik (STPI)	▶ Mr. Swapnil Jalundhwala (Adani Aerospace) ▶ Mr. Rahul Jain- Geo Matrix ▶ Wing Commander Rajesh Sharma (1 Bengal Air Squadron NCC) ▶ Mr. Aakash Sinha (Omnipresent Tech)
Drone Competition	
▶ 360 Degree Solution ▶ FPV Drone Race ▶ Dhanvantri	▶ FPV Drone Race ▶ Scan & Lift the Payload ▶ Dhanvantri

12. Organization Presentations

12.1. DGCA

The Directorate General of Civil Aviation is the regulatory body in the field of Civil Aviation primarily dealing with safety issues. It is responsible for regulation of air transport services to/from/within India and for enforcement of civil air regulations, air safety and airworthiness standards. It also co-ordinates all regulatory functions with International Civil Aviation Organisation.

नागर विमानन महानिदेशालय
DIRECTORATE GENERAL OF CIVIL AVIATION

DGCA was represented by **Mr. Hillol Biswas**. He started the presentation with thanking the authorities to organise such an important event. His focus of the presentation was Current Regulatory situation for Drone operations in India. During his presentation, he covered related regulations, and recent changes given the dynamic nature of drone applications its potential threats. He also talked about clearances, regulatory compliances related to imports, training and operations. He also touched upon the direction in which department is thinking about counter drone and related regulations. Please refer section below, which highlights key aspects of this session.

12.1.1. Rule 5A of Aircraft Rules, 1937

- Empowers DGCA to issue directions in the interests of the security of India or for securing the safety of aircraft operations.
- Considering potential threats posed by drones, on 7th October 2014, DGCA issued a Public notice restricting non-government agency, organization, or an individual from launching UAS in Indian Civil Airspace for any purpose whatsoever.
- Vide GSR No. 1358 dated 1st November 2017, Rule 15A - Operation of Remotely Piloted Aircraft System was inserted in Aircraft Rules 1937.
- On 1st November 2017, the draft CAR Section 3, Series X, Part I on civil RPAS was uploaded in DGCA website for public comments.
- The regulation on remotely piloted aircraft system (RPAS) finalised after extensive consultations with various stakeholders and issued on 27th August 2018 and this regulation became effective from 1st December 2018.
- Operations of RPAS to be enabled through Digital Sky Platform.

12.1.2. Regulations Cover

- Categorization of Remote Pilot Aircraft (RPA)
- Issuance of Unique Identification Number (UIN) for RPA weighing more than 250 grams
- Issuance/Renewal of Unmanned Aircraft Operator Permit (UAOP)
- Security Clearance
- Remote Pilot training requirements
- Equipment Requirements
- Operational/ Procedural Requirements
- No Drone Zones
- Manufacturing standards
- Enforcement actions

12.1.3. Applicability of CAR

- CAR Section 3, Series X, Part I is applicable to Civil Remotely Piloted Aircraft Systems, which are Remotely Piloted from a Remote Pilot Station.
- Model and Autonomous aircraft are not part of the regulations.

12.1.4. Import Clearance

- Currently, Drones/ UAV/ UAS/ RPAS are restricted items and requires import clearance from DGCA and import license from DGFT.
- RPA weighing above 250 gm (i.e. Nano category RPA exempted from Import Clearance from DGCA.

- All drones should be No-Permission No Take-off (NPNT) compliant for issuing import clearance.

12.1.5. Operating requirements

All RPA operations are restricted today only, within Visual Line of Sight (VLOS), subject to following conditions:

- During daylight (between sunrise and sunset).
- In Visual Meteorological Conditions (VMC) with a minimum ground visibility of 5 km and cloud ceiling not less than 1500 feet (450 m).
- Surface winds of not more than 10 knots or as specified by the manufacturer.
- No precipitation (rain, hail or snow) or thunderstorm activities, or exceeding those specified by the manufacturer
- No person shall act as a remote pilot for more than one RPA operation at a time.
- RPA shall, at all times, give way to manned aircraft.
- RPA shall not discharge or drop substances unless specially cleared and mentioned in UAOP.
- RPA shall not transport any hazardous material such as explosives or animal or human payload.
- RPA operator/ remote pilot shall be liable to ensure that privacy norms of any entity are not compromised in any manner.

12.1.6. No Drone Zone

- 5 km from the perimeter of airports at Mumbai, Delhi, Chennai, Kolkata, Bengaluru, and Hyderabad.
- 3 km from the perimeter of any civil, private or defence airports.
- Within 25 km from international border (including LoC, LAC and AGPL).
- Beyond 500 m (horizontal) into sea from coastline.
- Within 3 km from perimeter of military installations/ facilities/ where military activities/ exercises are being carried out.
- Within 5 km radius from Vijay Chowk in Delhi.
- Within 3 km radius of State Secretariat Complex in State Capitals.
- Within 2 km from the perimeter of strategic locations/vital and military installations

12.1.7. Test Sites

- To encourage new technology, Indian Organizations/ Institutions involved in R&D related activity of RPAS shall use the 23 test sites listed in the CAR for testing/ demonstration purpose.
- The organizations, may alternatively utilize unused airstrips or Government educational institutions campus, provided adequate safety precautions are in place.

12.1.8. Counter Drones

- Ideally, DGCA is handling tasks akin to pro-drone system. Of course, basic equipment, systems, and features of manufacture broadly include location broadcast of the drone, identification, Detect-And-Avoid (DAA), etc.
- Bureau of Civil Aviation Security (BCAS) is spearheading the counter-drone system.
- Ministry of Civil Aviation vide its order dated 20th April 2018 constituted a committee under the chairmanship of Director General-BCAS.
- The said committee had members from MHA, DRDO, NTRO, DGCA, AAI, CISF, etc.
- The task was to prepare and finalize basic requirements and technical specifications for counter-drones' technologies/ solutions for surveillance, detection and neutralization of rogue drones.
- The further updates (if any) on this issue can be provided by our esteemed colleagues from BCAS.

12.2. IIO Technologies

IIO Technologies is a Bengaluru based startup and pioneer in providing solutions in Counter UAV Solutions & Tactical Drones space. The organization has built solutions primarily focusing the needs of Army, Airforce, Navy, Airports, Critical infrastructure, Law & order agencies, & VIP protections.

IIO Technologies was represented by its CEO **Mr. Sunny Sharma.** He showcased his counter drone technology product demo. He talked about application of drones, specifically related to security & safety challenges posed by it. He quoted examples of disturbance on airports due to drone, and past drone attacks on world leaders and in large gathering. He suggested that counter drone technologies developed by IIO Technologies can provide a comprehensive solution in dealing with such challenging situations. During the presentation he also stressed that besides technology, drone users must also be educated as many users can trigger a catastrophe unintentionally. Counter drone technologies are meant for criminals whereas careless users need to be educated. He suggested to use combination of radio frequency – to detect drones – and jamming techniques to deal with rouge drones.

Please refer section below which highlights IIO's counter drone solutions & recommendations -

12.2.1. Detection Solution

Radio Frequency (RF)
Identifies the presence of drones by scanning for the frequencies on which most drones are known to operate. Algorithms pick out and geo-locate RF-emitting devices in the area that are likely to be drones.

Electro Optical (EO)
Detects drones based on their visual signature

Infra-Red (IR)
Detects drones based on their heat signature.

12.2.2. Interdiction Solution

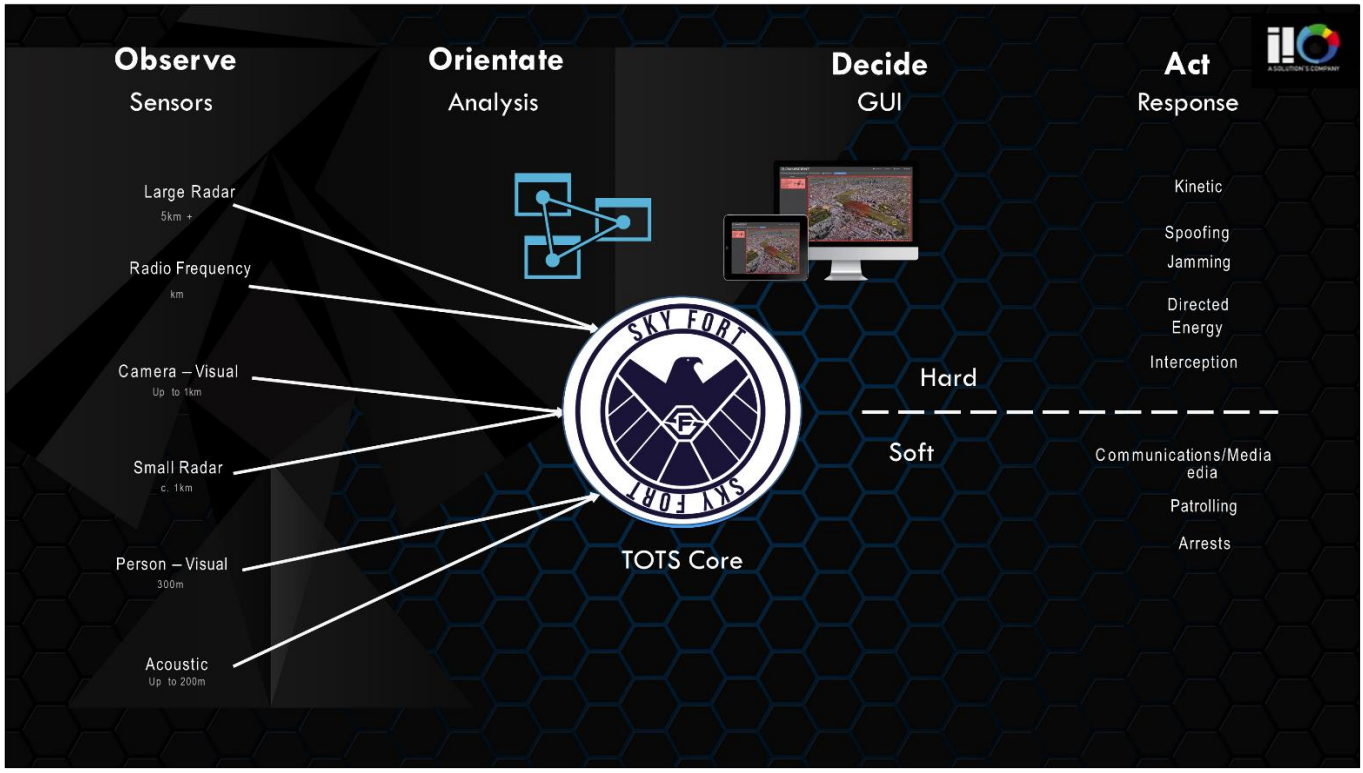
RF C2 & GNSS Jamming - Disrupts the radio frequency link between the drone and its operator by generating large volumes of RF output. Once the RF-GPS link, which can include Wi-Fi links, is severed, a drone will either descend to the ground or initiate a "return to home" manoeuvre. Here are the solution highlights –

- Long Range Detection
- Open Architecture / Sensor Fusion
- Long Range Narrow Jamming
  Sensor Agnostic
- AI & ML Enabled Anomaly Detection
  GPU-Powered
- ATC Asterix Single Window
- 3D Unified View
- UAV Flight Prediction
- Friend & Foe Detection
- Signal of Interest Detection
- Proprietary Remote ID Integration
- 50Km Range

IIO Technologies has designed the system to integrate solutions as per user requirement. Here is the aadditional 3rd Party Integrations -

- Kinetic
- Jamming
- Spoofing
- Directed Energy
- Interception
- Ballistic

SKYFORT Solution



12.2.3. Recommendation

Here are the key recommendations suggested –

- Ministry of Civil Aviation has been extremely encouraging by initiating OAN & DAN registration. Allowing users to self-declare.
- Community Watch – Toll Free Number for drone sighting and reporting
- Awareness Media Campaign - No Drone Fly Zone Sign Boards
- Education/Training - inform careless and clueless UAS operators about the risks and penalties associated with unauthorized or unsafe UAS operations
- Enforcement - Laws prohibiting UAS operations in restricted areas, including airports, must be strictly enforced with heavy penalties and legal actions.

12.3. Milkor

Milkor is a South African defence company which was established in 1981 as an original designer and manufacturer of the handheld 40mm Multiple Grenade Launcher weapon systems. Milkor has supplied in excess of 64,000 weapons to more than 67 countries worldwide to military and law enforcement ranging from lethal to less-lethal.

Milkor's vision is to be a developer and supplier of a wide range of defence and security products. Our current product range includes the design and build of armoured vehicles, patrol boats, unmanned aerial systems, cyber security solutions, and weapon systems.

Milkor has recently opened its new office in New Delhi in India and is keen to expand its presence here. Milkor India was represented by its Manging Director **Mr. Luciano Jardim**.

During his presentation, Mr. Luciano showcased Milkor's capabilities, its forward thinking mindset, and talked about how Milkor can help India grow as a next major exporter in UAV systems. He also mentioned that he is looking forward to establishing a manufacturing hub for Milkor's products in India. Please refer section below, which highlights the firms capabilities and key solutions.

12.3.1. CAPABILITIES

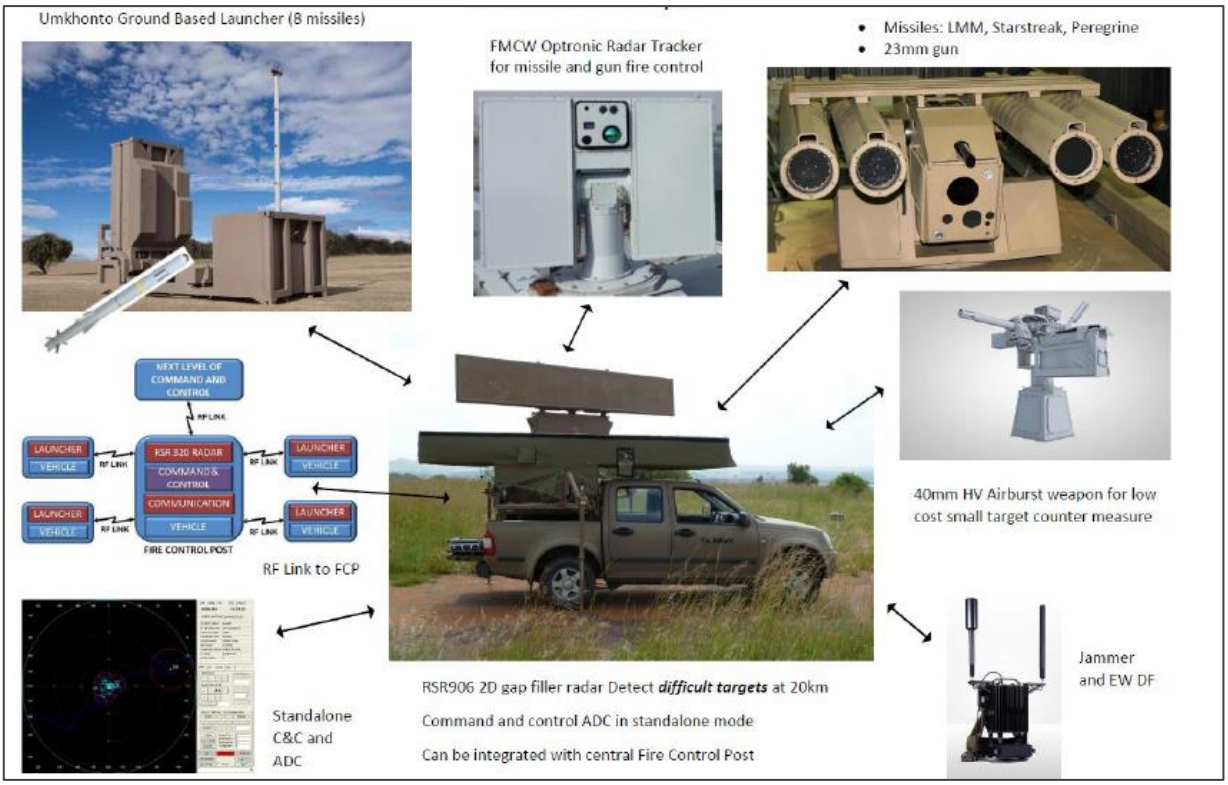
- LAND SYSTEMS We offer world class land capabilities tailored to the different specifications and needs of each project. Specializing, but not limited to, providing Armoured Personnel Carriers (APCs), weapon stations, intelligence solutions.
- NAVAL SYSTEMS Milkor designs, manufactures and offers after sales service for Naval and Commercial requirements.
- AIR SYSTEMS Milkor's aeronautical engineers' experience and ingenuity have allowed for a UAV combined with a UAS that utilises industry leading technologies that offers solutions to suit your specific needs of Aeronautic forces worldwide.
- CYBER SECURITY We use highly integrated technology to detect the smallest breach in cyber activity within an organisation, regarding both internal and external threats. The software solutions that we offer are recognised worldwide and our specialist security team works closely with our customers to establish and maintain information security by identifying, managing and controlling risks.
- WEAPON SYSTEMS Milkor is internationally known for its 40mm Multiple Grenade Launchers (MGLs) which are used in more than 67 countries worldwide. The production of these weapons will always be an integral part of our company's success.

12.3.2. Products

MAN PACKABLE AERIAL FIRE SUPPORT

- Tri-tilt rotor airframe
- Nominal Take off mass – 6,5kg
- Endurance – 22 minutes
- Communication range – 3-5km
- Construction – Carbon fibre and aluminium
- Firing – 1 x 40 mm HE rounds with a 5m kill dispersion
- Handle recoil

Scalable VSHORAD SYSTEM



12.4. IdeaForge Technology

ideaForge is India's largest manufacturer of drones for defence, homeland security and industrial applications. Founded in 2007 by IIT-Bombay alumni, the organization has a consistent market share of over 90% in Security & Surveillance segment. The organization is a licensed manufacturer of UAVs approved by Ministry of Defence (MoD).

ideaForge is the first organization to indigenously develop and manufacture Vertical Take-off and Landing (VTOL) UAVs in India. It is the pioneer in the UAV industry in India and has multiple IPs to its credit, including the then world's lightest autopilot in 2009. Currently, the organization has deployed over 700 systems and has trained over 1300 pilots in services including Indian Army, Navy, Air-Force, all CAPFs (CRPF, BSF, NSG, etc), State Police Forces, Indian Railways, TAFE, NTPC, DRDO, L&T and many more.

ideaForge was represented by **Mr. Aditya Thapliyal.** During the interaction, he presented the capabilities of ideaForge, and specifically focused his discussion around forest management using drone. Please refer below, which highlights key aspects of this area and relevant solutions.

12.4.1. Global Impact of forest fire & India Scenario

32.5 billion tons of Green House Gases got released each year and 20 percent of this is released by forest fire. Green House Gases released by forest fires each year will increase by 30% and may partially cause global temperatures rise by more than 3°C by 2100. The world also faced an economic loss of USD 52.3 billion from 1987-2013.

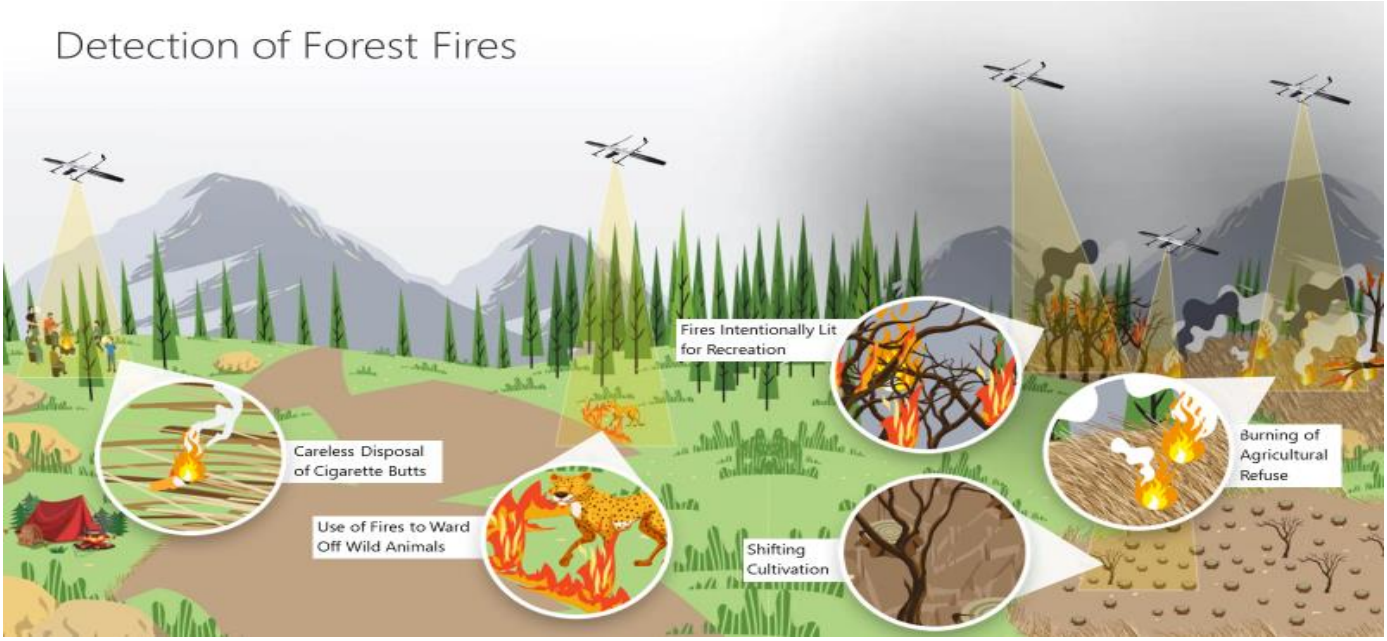
30,300 Forest fires are reported in India In 2019 and 95 percent of it is not due to natural causes but due to human activities. Only 4% of forest fires are due to natural causes. The rest are caused by human activities.

12.4.2. Why Uttarakhand Needs Drones

Uttarakhand Forest Area- Total Area of 24,295 sq. km which is 45.43 % and we can't afford to lose this forest cover. To mitigate the risk of forest fire, the Uttarakhand government has formed a Drone task force.

The only way to mitigate this risk is - **Detection Identification Mitigation**

Forest fires can be contained if detected in about15 minutes using Drones



12.4.3. Advantages of ideaForge Solution

Up to **25x** More Area Coverage	Up to **45%** Reduction in Emergency Response Time	Up to **40%** Reduction in Patrolling Costs	Up to **85%** Reduction In Deployment Time as Compared to a Helicopter

12.4.4. IdeaForge Product portfolio

NINJA UAV
Lightweight Scout

- **20-25 min** Flight Time
- **Over 2 km** Operational Range
- **15+ MP** HD Imaging Payload
- **Up to 2 kg** Take-off Weight

Q Series UAV
Enterprise Specialist

- **Over 40 min** Flight Time
- **Over 4 km** Operational Range
- PPK Enabled Survey Grade Mapping
- **Up to 3.5 kg** Take-off Weight

NETRA Pro UAV
Versatile Workhorse

- **Over 30 min** Flight Time
- **Over 4 km** Operational Range
- Supply Drops/ LiDAR
- **IP53 Rated** for Dust and Water Resistance

NETRA V Series UAV
Endurance Champion

- **Over 60 min** Flight Time
- **Over 5 km** Operational Range
- **HD 10x Optical Zoom** 3-Axis Stabilized
- **IP53 Rated** for Dust and Water Resistance

SWITCH UAV
All Terrain Dominator

- **Over 120 min** Flight Time
- **Up to 15 km** Operational Range
- **VTOL** Capability
- **IP53 Rated** for Dust and Water Resistance

12.4.5. Creating Impact – key success stories

Nepal Earthquake SAR Operations

- NETRA UAVs were deployed for site surveillance in Kathmandu and neighbouring areas and helped the rescuers assess the extent of damage to the infrastructure in marked areas and how to access them to implement search-and-rescue operations.

Uttarakhand Floods SAR Operations

- The National Disaster Relief Force (NDRF) deployed 4 NETRA UAVs to assist the rescuers in their missions. The UAVs were used to scan the landscape including other inaccessible or blindside zones to identify survivors and send their exact location to the mission control room.
- Reconnaissance at least 50 areas including 20 that were inaccessible to the rescuers for damage assessment.

12.5. STPI

Software Technology Parks of India is an Autonomous Society set up by the Ministry of Electronics and Information Technology (MeitY), Government of India in 1991, with the objective of encouraging, promoting, and boosting the Software Exports from India. Software Technology Parks of India maintains internal engineering resources to provide consulting, training and implementation services. Services cover Network Design, System Integration, Installation, Operations and maintenance of application networks and facilities in varied areas. Process development is based on the Quality Management System. Software Technology Parks of India centre also adhere to ISO 9001 certification.

India has earned itself a reputation of an Information Technology superpower. Software Technology Parks of India has played a seminal role in accomplishing this status. Today, Software Technology Parks of India across the country is synonymous to excellent Infrastructure and Statutory support aimed at furthering growth of Information Technology in the country.

STPI was represented by **Mr. Ganesh Naik**. He talked about the role of STPI in India and how it is working towards strengthening the capabilities in emerging technology area by forming Centre of Excellences. Please refer section below which highlights key aspects of his presentation.

12.5.1. Centre of Excellence

STPI Launched CoE service in India, a pioneer step that enabled 156 STPI members to avail 1.8 lakh sq ft space across the country

- Develop one-stop vibrant platform and ecosystem for 360-degree support to start-up's,
- Build an ecosystem for R&D and Products innovation in emerging technology
- Nurture tech start-up to develop indigenous products and create IPR
- Augment India's leadership position in emerging technology
- Bolster economy through start-up revolution
- Boost employment generation

STPI provides holistic support to incubates

- **Collaborative** -Bringing all stakeholders on a single platform
- **Comprehensive** All-inclusive support system from ideation to IP creation
- **Convenience** Easier for start-ups to adapt the ecosystem and scale up
- **Competitiveness** Support from Proof-of-Concept to Go-To-Market to become competitive
- **Cost-effective** Affordable for all

12.5.2. COE Model & Guidelines

12.5.3. Start-ups in Drone Solutions

Currently, just about 40 drone start-ups are active in India. Over 90% of them are start-up's that do not have the luxury of burning cash on trials and experimental runs in this regulatory uncertainty. Among these, many are flying high in India's digital sky.

12.5.4. Dehradun & Drone Lab

- ITDA, Govt of Uttarakhand has already established a basic Drone lab in joint venture with NTRO in 2018 at IT Park, Dehradun.
- Uttarakhand is a hilly state and has a difficult terrain which is prone to natural disasters.
- Huge scope for utilization of Drone Technology in the Himalayan state during disaster.
- STPI coming up with its 2nd state-of-art Centre in Dehradun

Prospective Services & Industry

Services	Sector/Industry
Surveying and Mapping.	Agriculture and Forestry
Inspection and Analysis	Emergency and Disaster relief
UAV Data Processing	Healthcare for remote areas
Aerial Photography	Construction & Maintenance industry
Project Monitoring	Anti-drone and security applications
Drone Video	Mining
Aerial Sports Broadcast	Road and Rail
Disaster Management	Solar
3D Reconstruction	Govt. Safety & Surveillance
Scientific Data Collection	

12.6. Matrix-Geo

Matrix-Geo Solutions is a multidiscipline, full-service photogrammetric mapping firm specializing in providing Photogrammetry, GIS - RS and LiDAR solutions. Matrix-Geo has provided Photogrammetric and Geospatial solutions including the data creation and product development solutions for medium to large scale enterprise and government agencies across the world. Matrix-Geo is known for dependability, accuracy and superior quality with a consistent track record in meeting technical requirements on-time and on-budget.

Global Experience and clients

- **International** -World Bank USA, 50+ International clients including 15+ Active & repeated business client
- **Government** – Railways, NHAI, PSUs (IRCON, RITES, DFCCIL, RVNL, NHAI, WAPCOS, NTPC), Municipality
- **Private** – ADANI, WELSPUN, DALMIA, LANCO, LnTECC, CEG, STUP, ATKINS, JYPEE

Matrix-Geo was presented by **Mr. Rahul Jain.** During the presentation, he showcased Matrix-Geo capabilities, its presence and solutions. He also talked about the various challenges faced by the drone industry and what possible governmental interventions can ease them out. Please refer the section below which highlights key aspects of his presentation.

12.6.1. Anti-Drone - A real requirement in present security scenarios in INDIA

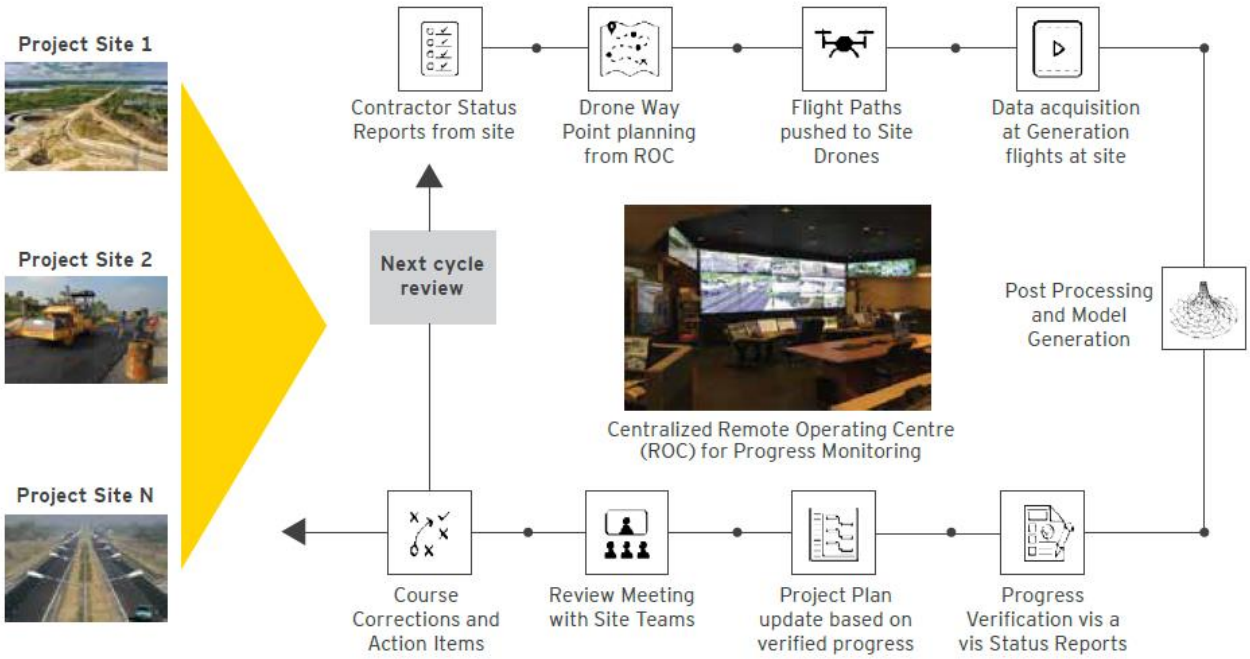
A new security threat – Drones weaken security at all sites, creating new risks and potential threats: Spying, contraband, smuggling, collision or even terrorist attacks

12.6.2. LAAMS: Low Altitude Aerial Monitoring / Mapping Systems (LAAMS)

The Low Altitude Aerial Monitoring / Mapping Systems (LAAMS) which use UAS (Unmanned Aerial system) with high quality camera to capture geo-referenced aerial images & videos and later process them using the **Photogrammetry technology** for measurement and create accurate topographical information's relevant for planning and monitoring quickly, safely & cost effectively.

- Equipped With Dual GNSS (Precise Position)
- Equipped with Calibrated Camera
- Equipped with Ground Control Station for flight planning

12.6.3. Life cycle of UAV projects



12.6.4. Our Innovation and Products

Scanfly Drone based LiDAR System

- Foldable Drone
- Very Light weight LiDAR sensor
- Capture 2 Cm Resolution Images from integrated Camera
- Accuracy of 2-5 CM
- Weight - <2 kg
- Endurance - 35 min
- Multi Return

Skyhov-1 DGPS Integrated Drone

- Foldable Drone that can fix in an overnighter
- DGPS (RTK/PPK) Integrated Drone
- Capture 1-2 Cm Resolution Images from Calibrated Camera
- Images with Accuracy of 2-5 CM
- Weight - <3 kg
- Endurance - 35 min
- Payload - 800 gm

Skynetra PPK Upgrade kit Convert DJI drones to Reliable survey grade drone

- Accuracy of 2-5cm RMSE
- Lower down your project cost by reducing GCP
- Fast geotagging of correct coordinates.
- Built in survey grade GPS, GLONASS, BeiDou, Galileo, SBAS
- Can be installed on any DJI Drones
- Camera calibration for Geomatics Applications

12.7. Adani Defence and Aerospace

India aspires to be amongst the top five nations of the world in defense and aerospace, by reducing its current dependence on military imports, by bolstering defense manufacturing, through indigenous private defense firms.

Adani Defense and Aerospace leads the Adani Group's foray into defense and aerospace manufacturing and in continuation of Adani's vision of nation building, we are keen to play an instrumental role in helping transform India into a destination for world class defense and aerospace manufacturing, aligned to the Make in India initiative.

Adani Defense and Aerospace will achieve this by:

- Focusing on platforms and technologies of critical importance to meet emerging security challenges
- Collaborating with committed global partners for transfer of technology and skills
- Helping grow the dynamic MSMEs to develop a sustainable defence ecosystem in India
- Focusing on capabilities critical for indigenisation including design, system integration, maintenance, and upgrades in India

Adani Defence and Aerospace was represented by **Mr. Swapnil Jalundhwala.** During the interaction, he presented Adani Defence & Aerospace capabilities, its future plans and how its activities are aligned with Make in India campaign. He also talked about the roadmap to develop the UAS ecosystem in the country.

12.7.1. A world class infrastructure for defence manufacturing

- Private aerodrome with 2.2 Km long airstrip
- Industrial ecosystem for manufacturing in SEZ
- Connectivity to Port, Rail and National highways
- Uninterrupted access to power, water and other utilities

12.7.2. Manufacturing Platforms & Technologies in India

Adani has set up an ecosystem of capabilities in India

- System and Avionics
- Carbon composites aerostructure and assembly
- Research and development
- Transmission gearbox and gear
- Metal aerostructure and assembly

Adani has a full spectrum of Unmanned Aerial Systems under Make in India.

12.7.3. Thor – Military UAS

Specifications

- Take-off weight: 40 Kg
- Endurance: 6 Hours
- Range: 100km
- Service Ceiling: 15,000 ft
- Wingspan: 4.7m

Key Features

- **Covert operation -** electrically propelled, extremely low acoustic and visual signatures
- **Versatility -** wide array of payloads including EO/IR, ELINT and COMINT
- **Advanced mission features -** handover between ground stations, Aerial hot swap, Remote video terminals
- **Simple to operate -** fully autonomous from takeoff to landing
- **Rapid deployment -** full deployment until takeoff in less than 20 min

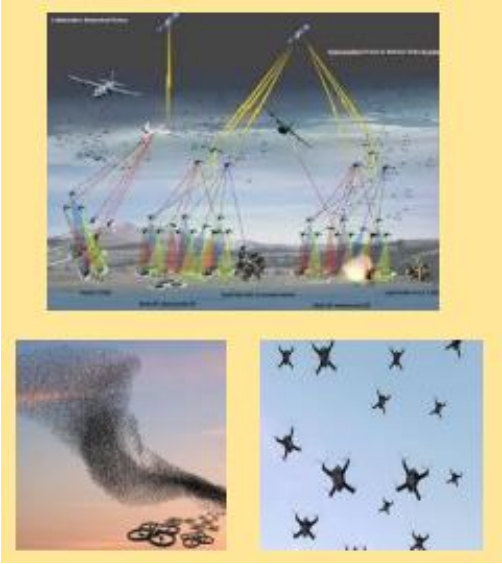
Global Success

- Newly designed and developed after the success of its earlier version Skylark I with 30 international customers
- Won one of the biggest programs in South East Asian Country
- Adani-Elbit JV has been contracted to manufacture 80 shipsets in 2 years for international customer

12.7.4. SWARM UAVs; Emerging Next Gen military weapon

The Concept Note

- Group of UAVs with an ability to fly together, change patterns, assemble & distribute with no intervention from the base
- Automation and Artificial Intelligence play a crucial role
- Technology is being perfected by US, Israel & China



Indigenization Efforts by India

- "Meher Baba" competition by IAF
- The winner gets an opportunity to further develop the concept with IAF
- 3 phases of competition completed. 5 teams have been shortlisted for the final phase. One of the most prestigious opportunity for the country

Adani's Participation

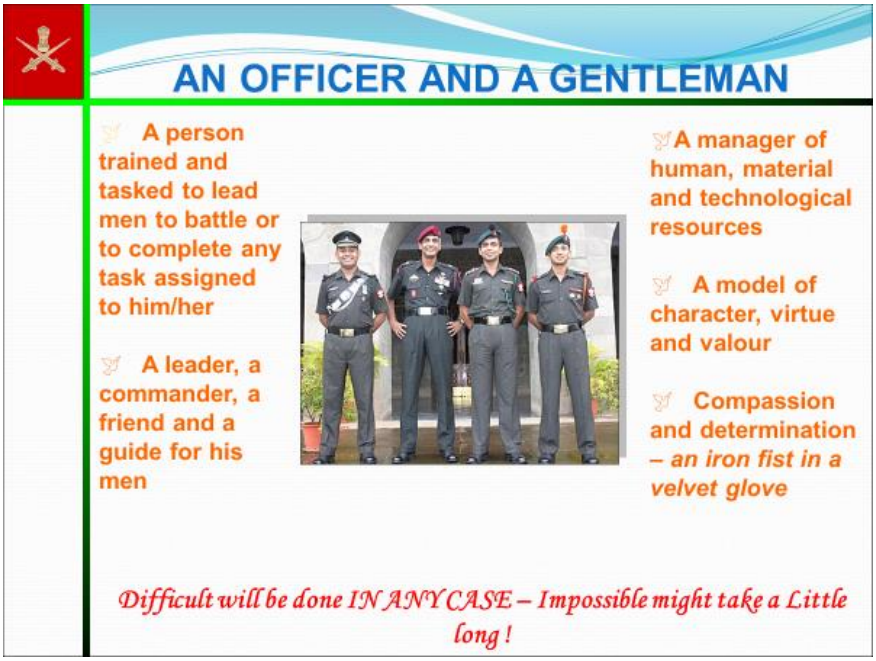
- Collaborated with one of the 5 shortlisted teams for joint development and manufacturing of the ambitious program
- Actively involved in design and development of technology modules required for proving the concept and the technology required by IAF. Successfully developed some of the critical ones

12.7.5. Ground Control System

- Tactical map display and system main control
- Multi-video display and payload control
- HOCAS (Hands on Control and Stick) advanced user interface
- Advanced HMI and automated operator support
- 1 Operator + 1 Commander
- Ruggedized, Transportable
- Dual UAV mission control

12.8. Bengal Air Squadron NCC

1 Bengal Air Squadron was represented by Wing Commander Rajesh Sharma (1 Bengal Air Squadron NCC). He started the presentation by congratulating everyone for making this event a huge success. While addressing the students he talked about the defence forces, their roles, and future prospects in case someone joins the defence.

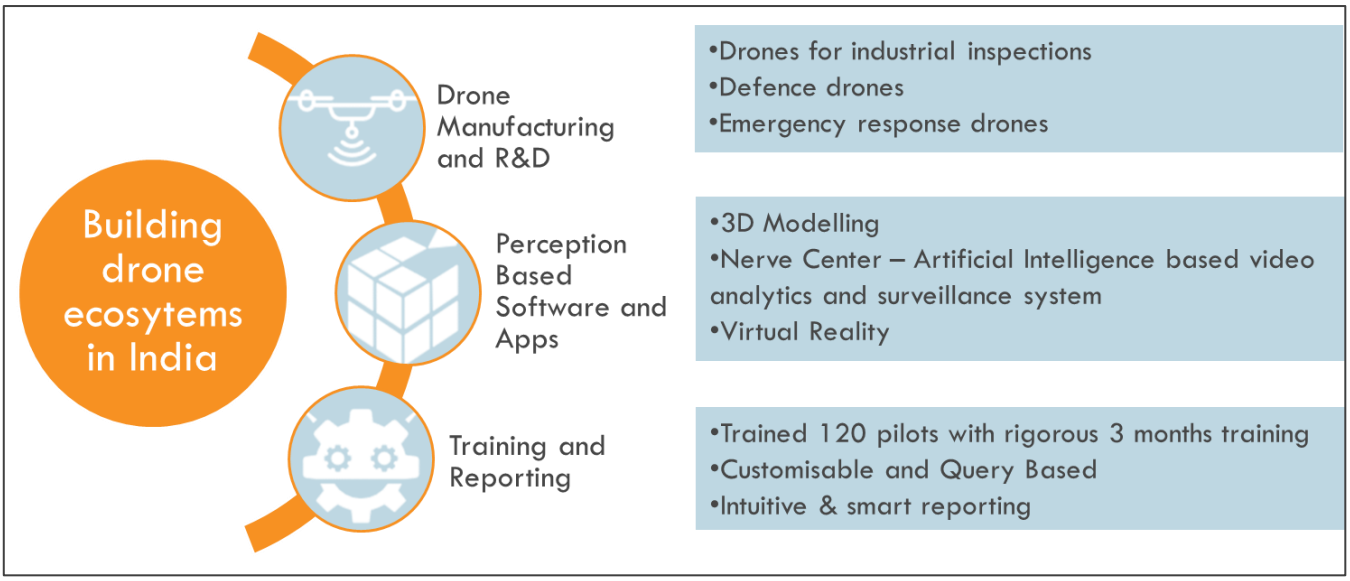


12.9. Omnipresent Robot Tech

Omnipresent Robot Tech is India's leading robotics, industrial UAV/drone and video analytics solutions provider. They build drones and robots for industrial inspections and defence and the software that drives them. Using computer vision, machine learning and virtual reality, they are able to provide our clients with unsurpassed visual analytics and actionable insights.

Omnipresent was represented by **Mr. Aakash Sinha.** He showcased Omnipresent during the event and talked about its capabilities and solutions. Please refer section below which highlights key portions of the presentation.

12.9.1. Building drone ecosystem in India



12.9.2. Services

Empowering large industrial companies to achieve optimized utilization of resources, enhanced productivity ratios and ensured implementation of safety standards by offering comprehensive drone, software or Artificial Intelligence based solutions.

- Site Inspections
- Encroachment Tracking
- Project Status Tracking
- Volumetric Calculations
- Surveying
- Drone Mapping

12.9.3. Industries

Wide-ranging solutions are capable of catering the needs of a diverse array of industries.

PORTS

OIL AND GAS

RAILWAYS

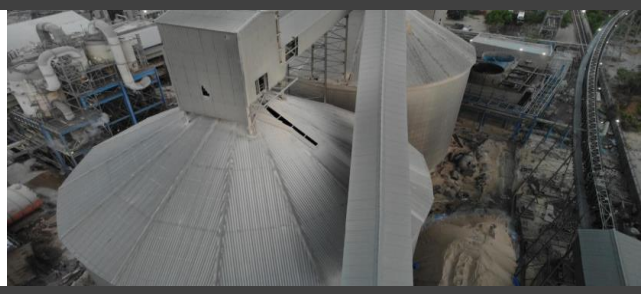
POWER/UTILITIES

CONSTRUCTION

12.9.4. Use Cases

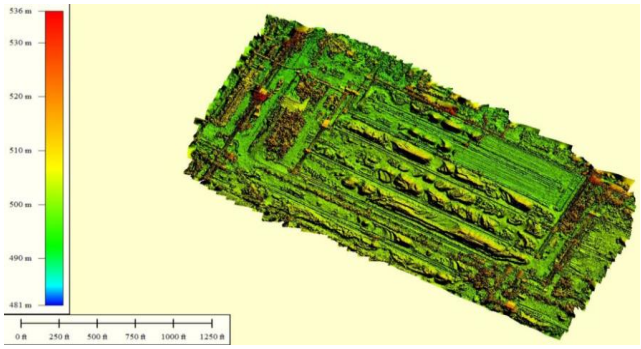
Inspection of Assets

- Inspection of Silos, Chimney, Storage roof, and cooling towers are done through drones as it is not humanly possible to inspect such places
- These inspections are like a diagnostic blood test for the plant, helping us avoid heart attacks(incidents)



Volume Calculation of Stock

- By using latest RTK GPS technology, we can get accurate volumes within 1% error
- All stockpiles such as coal, coke, limestone, bauxite, etc can be measured
- We can get accurate GL levels with manual digging/RTK GPS to measure GL
- Can be performed every month to assess stock accurately
- Technology deployed at multiple locations across HINDALCO, JSW, RIL and Ultratech cements.

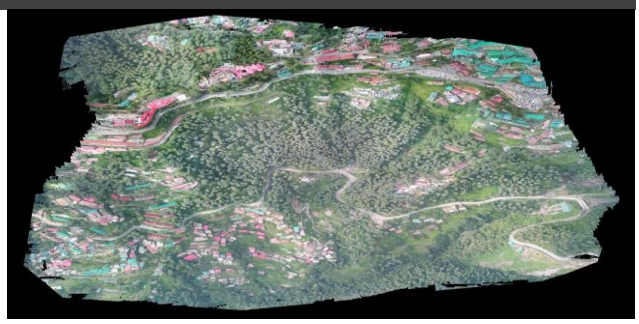


Plant Surveillance/Safety

- Get live video feed from the drone to HQ / control room
- Operate over a secured network
- Detect human activity / vehicles and count them
- Detect trespassing / theft
- Quickly deployed during emergency/fire
- Permanently deployed for in use in RIL Jamnagar for over a year

Mapping and 3D Model Generation

- Map accurately the entire plant/township/city.
- Calculate area, distance, height of any structure for all properties.
- Get layout for the entire plant and cities for planning and drainage assessment.
- We have mapped over 250 sq.km. of area in Amaravati, over 1000 sq.km. across Karnataka and another 1000 sq.km. In Maharashtra.



Automation in Mapping

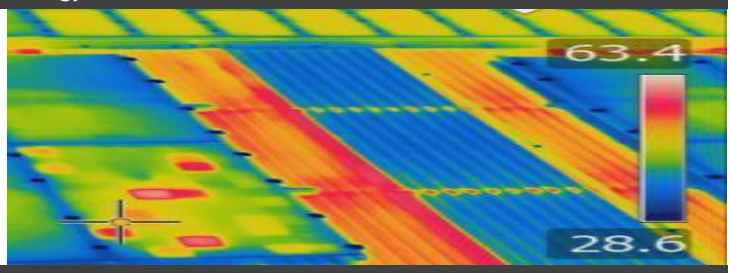
- We have been able to automate the most time-consuming activity of mapping – Cadastral map generation.
- Digitisation of aerial maps took months of manual work for large areas, but we have been able to complete it within a matter minutes with our in-house developed AI digitisation tool

Power/Tower Inspection

- Analytics based quick reporting with important parameters, facilitating decision making effectively
- Significant cost reduction compared to conventional methods
- Successfully served telecom majors like Bharti Infratel
- Improved quality due to high resolution images and videos from optimum perspectives
- Increased safety for inspection staff

Solar Energy

- Reduce measurement time from days to minutes
- Ensure that equipment is in the optimal position for maximum production
- Detect underperforming panels
- Eliminate measurement errors
- Safe alternatives for manual checks in harsh conditions



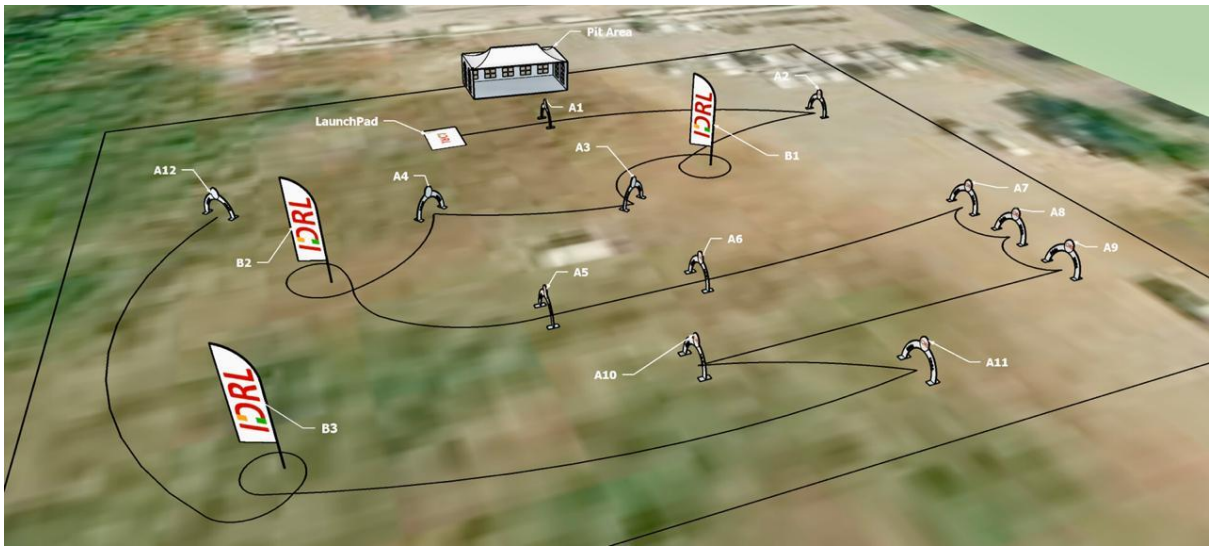
Railway Track/Bridge Inspection

- Tracks, tunnels, and bridges inspection can be done more effectively with very less manpower
- Government Initiative: Survey of 70,000 km railway track
- Less time consumption offers enough time for required action
- Conducted inspection of over 1000 km of railway tracks

13. DRONE COMPETITIONS

13.1. Drone FPV Racing

Drone Racing is the next frontier of high-speed racing sport that is bringing drones to the masses with an excitement and innovation unlike anything seen before. Drone Racers fly their drones at more than 170 km/hr through a live video feed sent from the drones directly to the pilot's goggles as they pass several gates and flags to complete a series of high speed laps, an adrenaline rushing feeling that makes the pilot feel like sitting in the drone's cockpit.



This sport has become very popular and widely adopted all over the world with an Olympics for the same that was recently conducted in Hong Kong where Indian drone racing pilots were not represented. There are many sponsors who are backing manufacturers of racing drones and their teams to send professional pilots all across the world to compete in multi-million-dollar races.

Problem Statement: Fly your RPA in first person view, through an obstacle course, in the minimum amount of time.

Rules:

- Maximum size of gate will be 1.45 m (height) x 2.6 m (width) approx.
- The RPA should be equipped with a on board live video transmission system.
- The pilot is allowed to fly by only looking at the live feed from the drone.
- There shall be qualifiers, semi-finals and finals
- No line of sight guidance is allowed.

Parameters to judge the winner: The minimum time to complete the obstacle course.

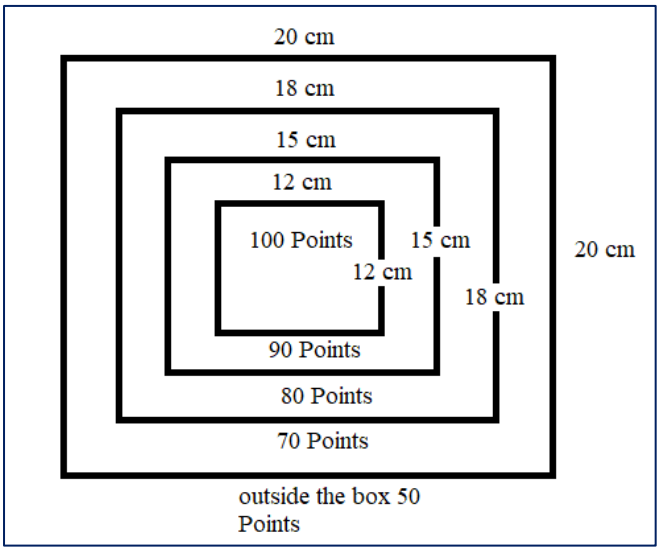
13.2. Scan and Lift the Payload

Problem Statement: Remotely verifying and adding QR/BARCODE data to a database by drones. Identifying the output and lifting the payload accordingly. It needs to lift a load of 100g of given dimension with minimum take-off weight and drop the payload at a desired location in minimum possible time. Adding a grabbing mechanism to lift the payload will ensure extra points

Rules

- Scan and store all the given QR/Barcodes through the drone at a given location away from the ground control station.
- Lift the dummy weight of 100g (Dimension: 5cm X 5cm X 5cm) attached with QR/Barcode code data provided by the organizers.
- The weights shall have a ring/hook to latch on. Any appendages/attachments shall not count as the weight of the payload.
- The payload may/may not be lifted by a grabbing mechanism.
- The participants cannot touch the payload in any way.

Parameters to judge the winner: Accuracy of load Drop:



*Fluffy surface to avoid bounce back of 100g weight.

Flight time:

S No	Flight Time (T)in sec	Points
1	T≤60	100
2	60<T≤90	80
3	90<T≤120	70
4	120<T≤150	60
5	150>T≤180	50
6	T>180	40

In case of a tie,

- The minimum time required to execute a given task will be calculated.
- The innovation/new idea used in design of UAV/Payload carrying Mechanics.
- After that, the maximum flight time of the drone shall be calculated if necessary.

NOTE: - If the two Teams secured the same points, the winner will be based on innovation/new idea used in design of UAV/Payload carrying Mechanics.

13.3. Dhanvantari

Problem Statement: RPA needs to carry and drop a medical kit of approximately 500g of given dimension, containing liquid in open cylindrical/rectangular tubes with a minimum take-off weight and drop the payload at a desired location in minimum possible time.

Rules:

- A sample containing box comprising one sample and medicine will be provided to the participating team. Volume and weight of the box is 8cm X 8cm X 8cm and 250 gm respectively. Any appendages/attachments shall not count as the weight of the payload.
- Sample is vulnerable to shocks and vibrations.
- The provision for suspension of the payload by a hook will be provided to the teams.
- The hook or any other payload method that the teams wish to use, should be arranged by the teams themselves.

Parameters to judge the winner:

1. Flight time:

S No.	Flight Time (T)in sec	Points
1	T≤30	100
2	30<T≤45	80
3	45<T≤60	70
4	60<T≤75	60
5	75<T≤90	50
6	T>90	40

2. Liquid level Drop:

S No.	Liquid Level (L) in mm	Points
1	L≤1mm	100
2	1mm>L≤2mm	80
3	2mm>L≤3mm	70
4	3mm>L≤4mm	60
5	4mm>L≤5mm	50
6	L>5mm	40

13.4. 360 Degree Solution

360 Degree Solution- Real Time Helicopter landing area detection

Abstract: Uttarakhand is blessed with natural beauty but at the same time Uttarakhand have to face natural disasters every year. During disasters, finding a suitable landing area for helicopter extraction is a key to the safe execution of the rescue missions. A number of requirements have to be met to land a helicopter. Upon the failure of landings, could cause the damage of resources, delay in objectives and even loss of lives. For tackling the disaster, SDRF performs various rescue operations. At the disaster site it is very difficult for the pilot to judge the safe area for helicopter landing for that AI base drone can be very useful in detecting the levelled surface for helicopter landing.

Problem Statement: Find software/hardware-based solution to locate an area to land a helicopter (Example: MI-17 V5, Chetak, etc.) in Real time. Onboard provision of the complete checklist of parameters used to verify the ability of the area to land the helicopter.

Parameters to judge the winner:

- Software/hardware Solution should be mountable on drones and helicopters.
- Software usability on drones.
- Onboard detection or on ground detection
- Latency of output result.
- Software stability and connectivity.
- Use of open source Hardware and software.

Deliverables: Software should be capable of:

- On board processing and calculation of all parameters required
- Flat land without any obstruction detection with area calculation
- Distance measurement from camera to landing point.
- detection of Lat/Long of landing site.

Abstract should have:

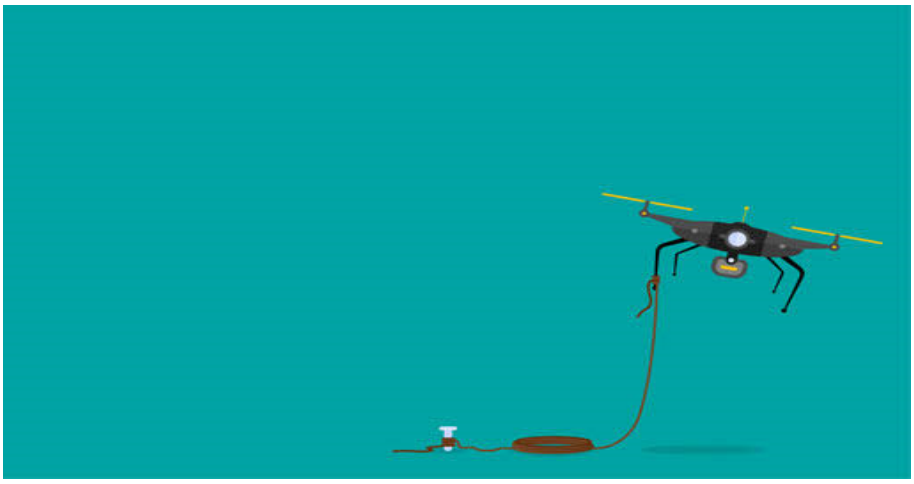
- Components used
- Method used

Abstract can have: Example work video

This event was cancelled since only two participants registered.

13.5. Tie the Knot

Abstract: Uttarakhand is blessed with natural beauty but at the same time Uttarakhand has to face natural disasters every year. During disasters, finding suitable means to supply food, medicines, and various other necessities to survivors across rivers and inaccessible areas at short distances upto 200-300 metres are a part of the various rescue operations that the SDRF performs. At the disaster site it is very difficult to fire grappling hooks on the other side of the land and hence a drone-based solution is required.



Problem Statement: An R.P.A. capable of carrying a rope upto 500g has to be towed and wound around a fixed object like trees, pillars, etc. The end of the rope, on the RPA has to be brought back to the take off point.

Venue: Arena 2 Timing: 19th Feb - 12:00 p.m. to 3:00 p.m.

Instructions:

- The number of seconds to come to the landing zone shall be noted as flight time.
- In case of a tie, the RPA with the minimum take-off weight shall be declared as the winner.
- The team/individual with minimum total point wins.
- Any failure to take-off from or landing on the landing zone will result in a disqualification.
- A successful placement of the loop in the pole is required.

Standard Operating Procedure (S.O.P.)

- The drone takes off from the take-off zone
- The drone places the loop in the pole
- The drone unties itself from the rope

- The drone lands on the take-off zone

13.6. Winners

- 1st Prize – Rs. One Lakh
- 2nd Prize- Rs. Fifty Thousand
- 3rd Prize – Rs. Twenty-Five Thousand

Winners - Dhanvantri Event	1. Mr. Ragul Kartikeyan 2. Mr. Janarthanan Jayachandran 3. Team Practice Hub	**Winners - Tie the Knot (360 Degree Solution) Event**	1. Mr. Ragul Kartikeyan 2. Mr. Vishal 3. Mr. Rakesh Sardar

Winners - Scan and & Lift the Payload Event	1. Mr. Ragul Kartikeyan 2. Mr. Janarthanan Jayachandran 3. Mr. Palash Paul	**Winners - Drone FPV Racing Event**	1. Mr. Devnath Sasidharan Nair 2. Mr. Aman Virani 3. Mr. Ritwik Nesargi

Winners - QUIZ	1. Ms. Abhilasha 2. Mr. Dheeresh 3. Mr. Vipulesh 4. Ms. Liya

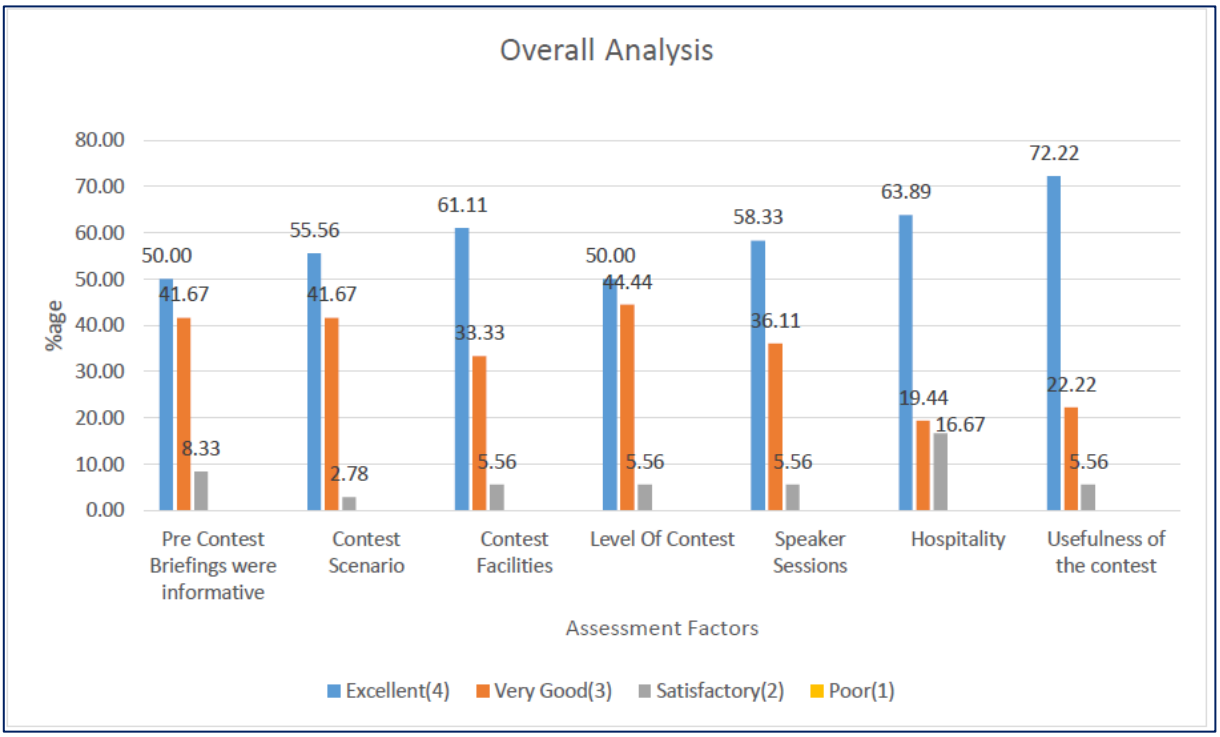
14. FEEDBACK ANALYSIS

Feedback questionnaire was designed to carry out a comprehensive survey of the event. Key elements of the questionnaire were:

1. Pre-Contest Briefings were informative and provided the necessary information.
2. Contest Scenario
3. Contest Facilities
4. Level of Contest
5. Speaker Sessions
6. Hospitality
7. Usefulness of the contest

The participants & visitors were asked to assess the overall performance and the extent to which the event objectives were met, the relevance and usefulness of the festival. The participants & visitors were asked to rate the experience on a scale of 1 to 4, with 4 indicating Excellent, 3 indicating Very Good, 2 indicating Satisfactory and 1 indicating poor.

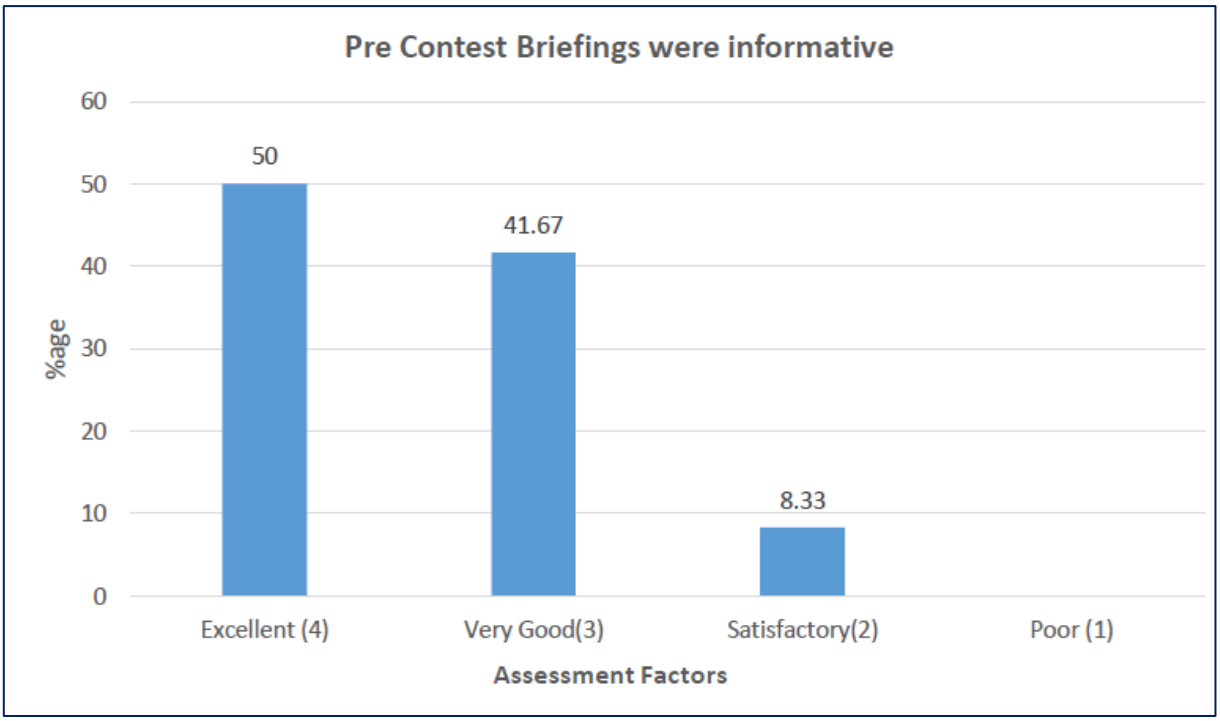
14.1. Participants' Feedback Analysis- Overall Analysis



Analysing the overall feedbacks given by the visitors we found that on average 58.74% of the visitors found the entire event perfectly organised and rated mostly assessments as excellent. 34.12% visitors on average rated the event very good, 7.14% visitors on average rated the organized festival as satisfactory.

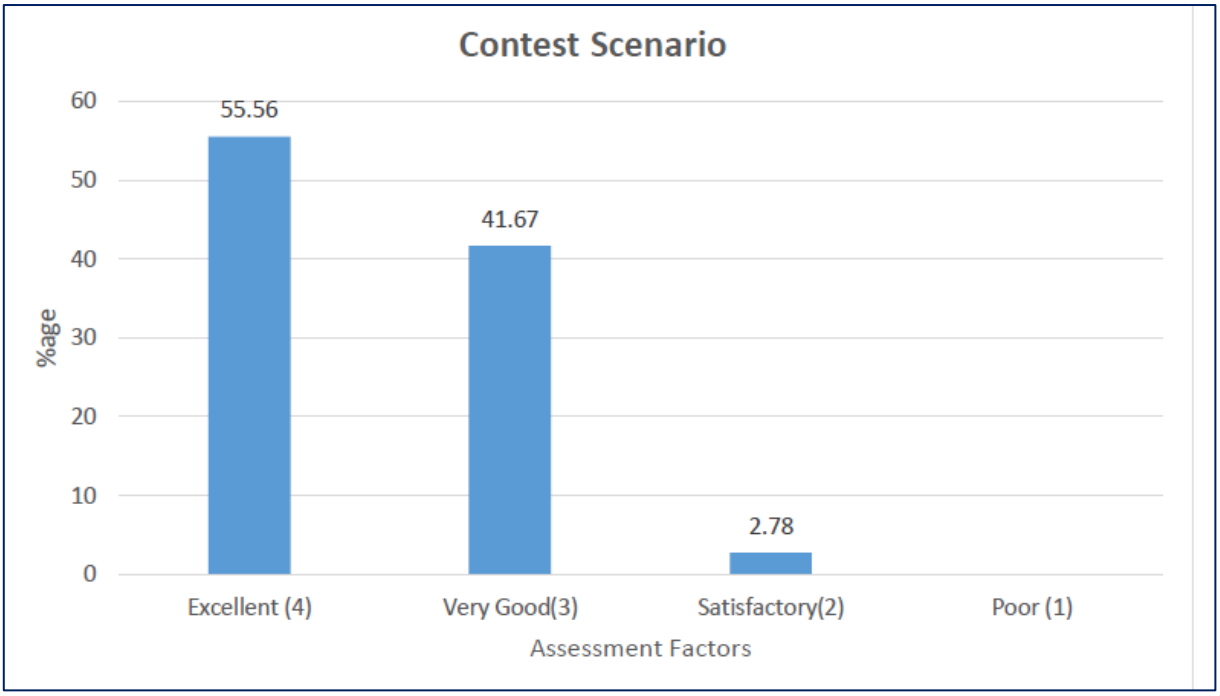
14.2. Feedback Analysis for Participants

14.2.1. Pre-Contest Briefings were informative and provided the necessary information



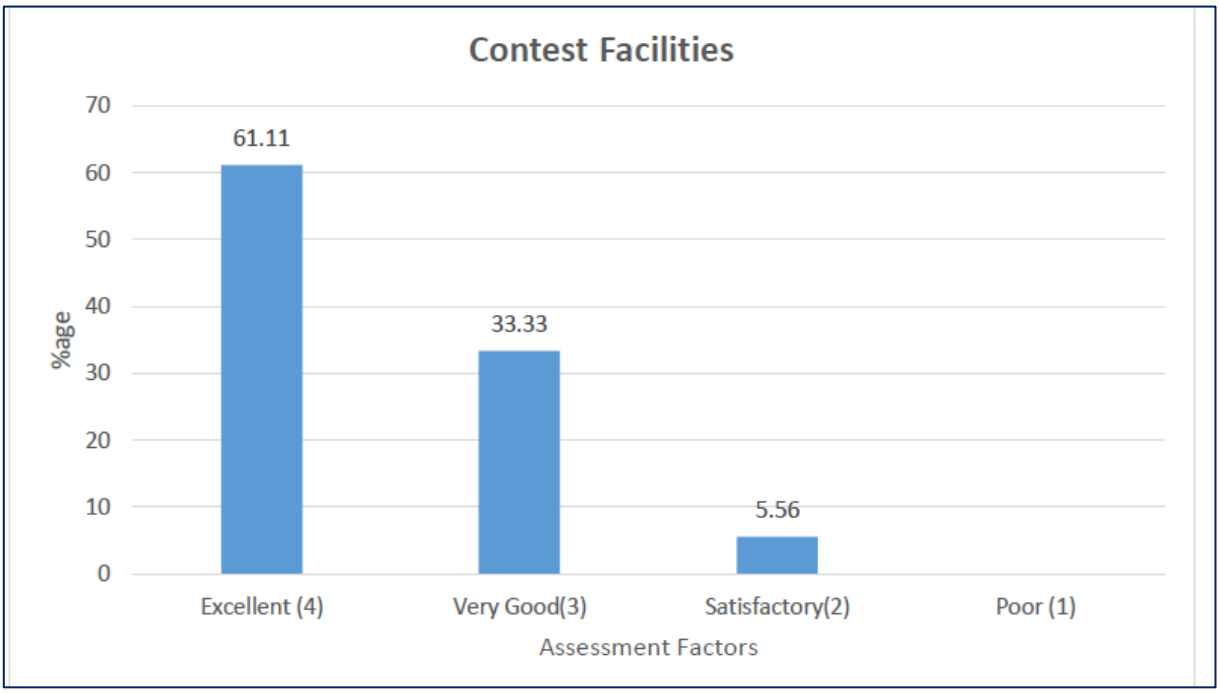
The participants had provided their valuable inputs for the pre-contest briefings were informative and provided the necessary information. As illustrated in graph 50% participants were of the opinion that the event was very well briefed. 41.67 % participant felt that briefing of the festival was very good. On the other hand 8.33% participants felt that the briefing was satisfactory.

14.2.2. Contest Scenario



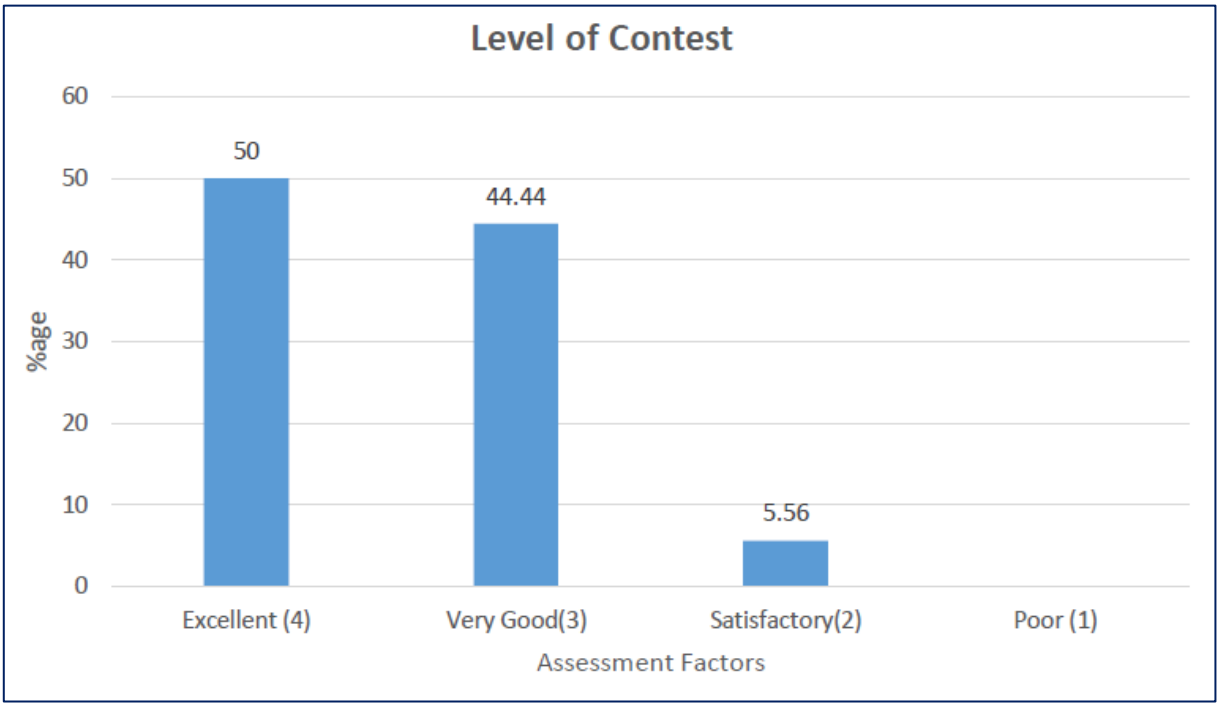
As per the feedback of the participants as mentioned in the above graph 55.56% participants had an opinion that the contests were Excellent. 41.67 % of participants felt that the contests were very good. On the other hand 2.78% participants felt that the contests were satisfactory.

14.2.3. Contest Facilities



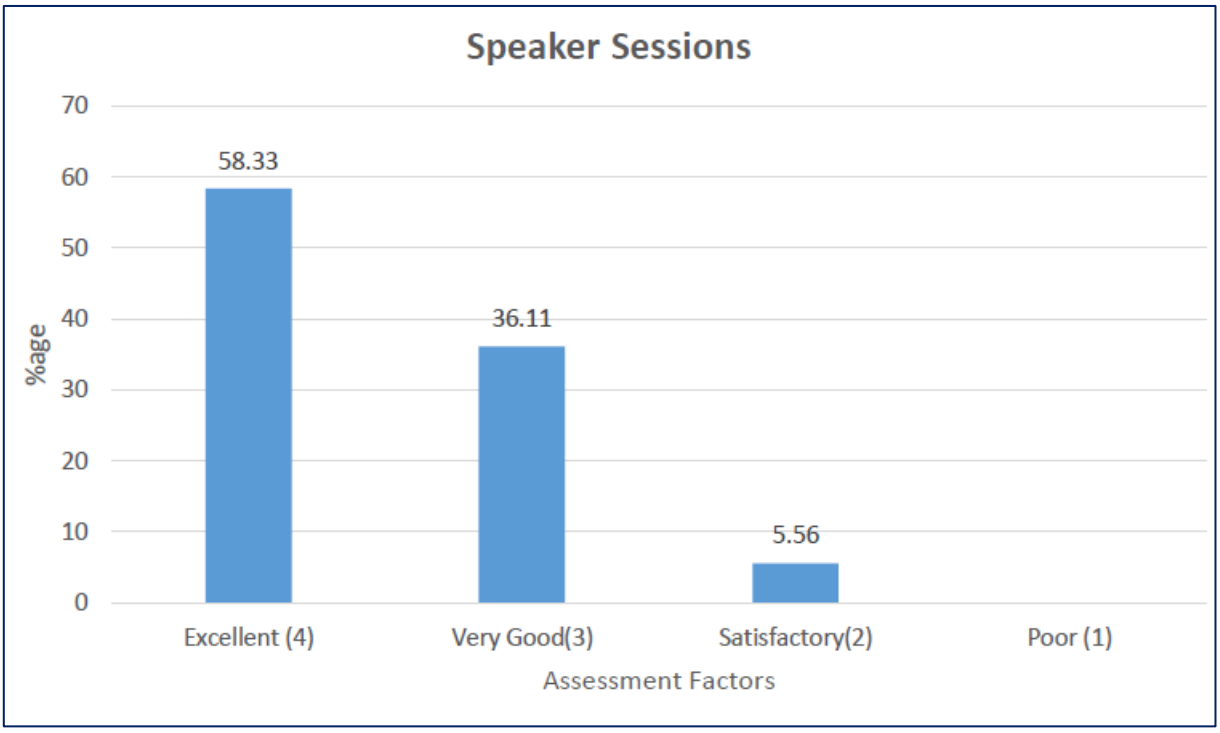
According to the above graph 61.11% of participants felt that facility provided in the Dronathon were excellent, while 33.33% found it very good and 5.56 % felt that the facilities were satisfactory.

14.2.4. Level of Contest



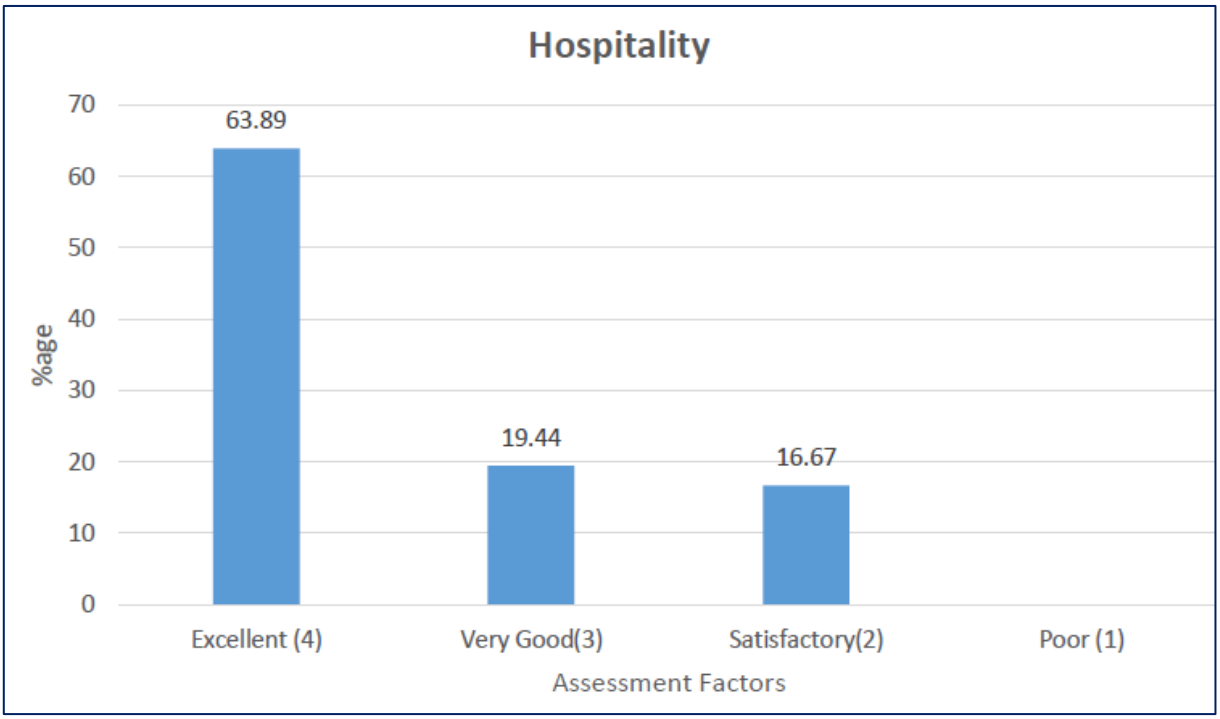
Analysing the feedback of participants in the contests as shown in the above graph 50% of participants felt that the level of contests in the Dronathon 2.0 were excellent, while 44.44% found it very good and 5.56 % felt that the level of contests was satisfactory.

14.2.5. Speaker Sessions



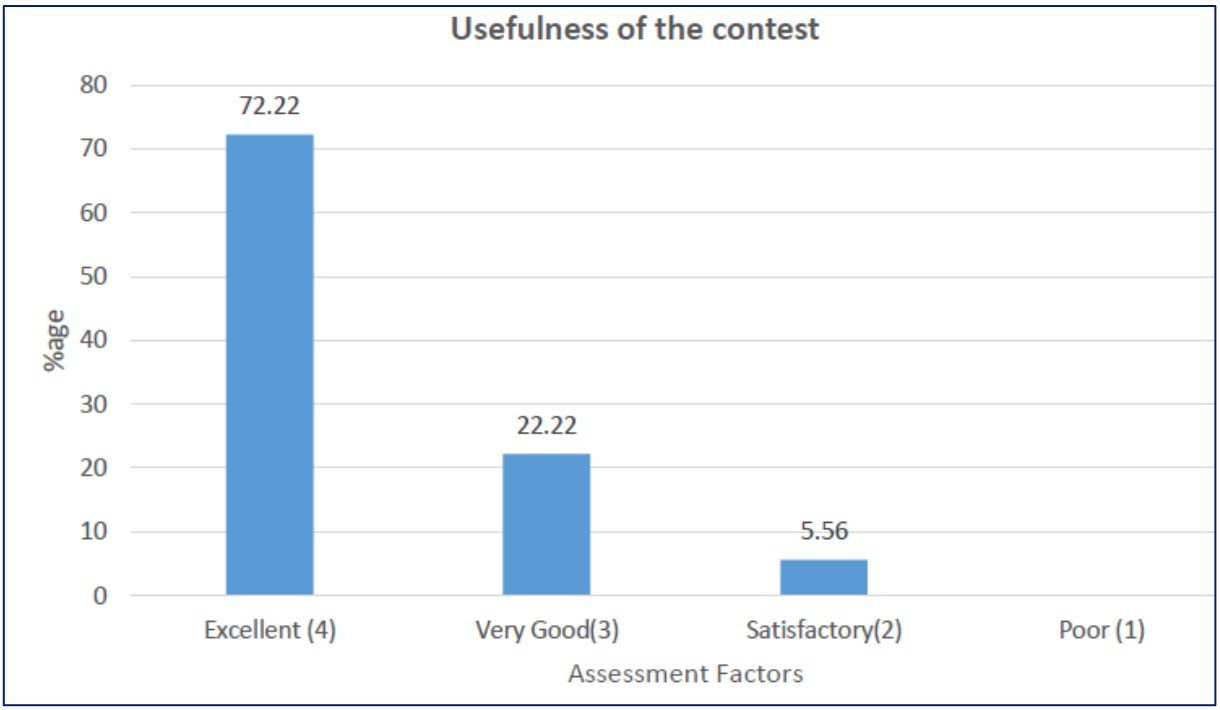
As per the above graph we can extract the details that 58.33% felt that the speakers' sessions were excellent. 36.11% felt that it was good and 5.56% found the sessions satisfactory.

14.2.6. Hospitality



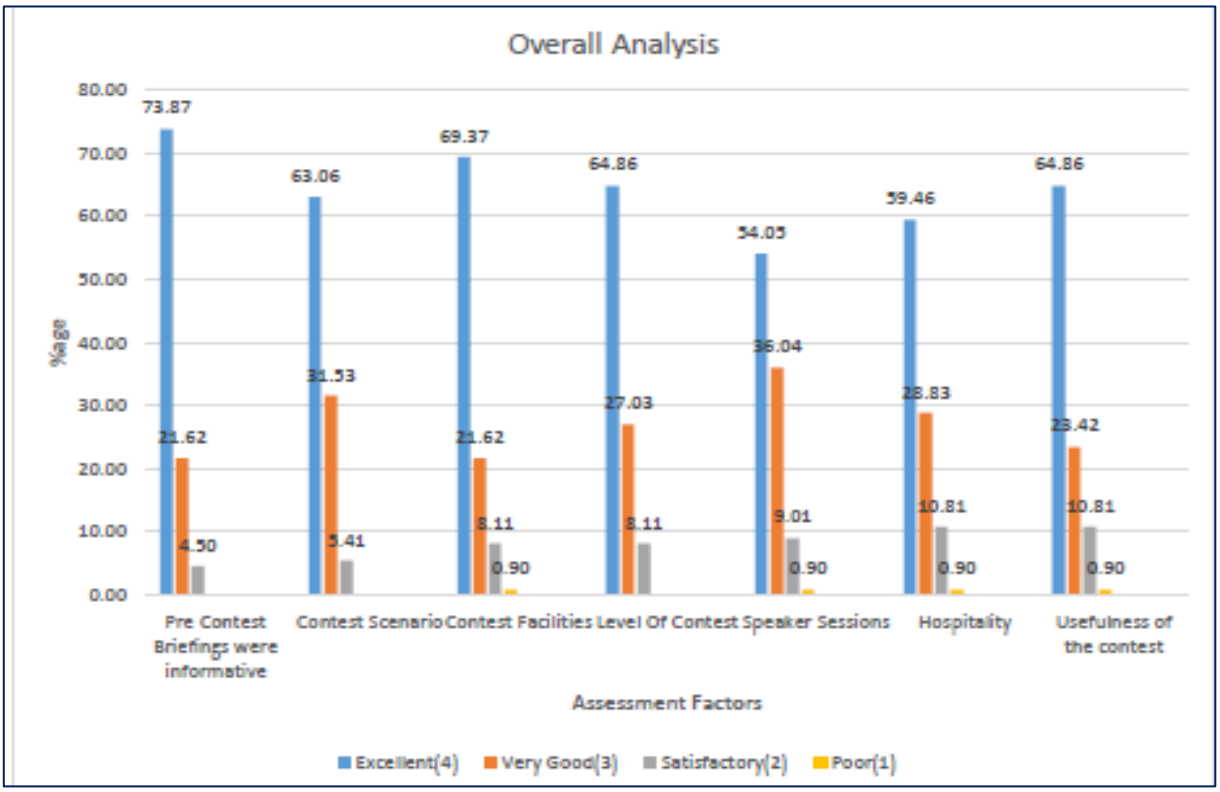
As per the feedback received by the speakers in Dronathon 2.0, 63.89% of them felt that hospitality received by them were Excellent, while 19.44% of speakers found it very good and 16.67 % felt that the hospitality was satisfactory.

14.2.7. Usefulness of the contest



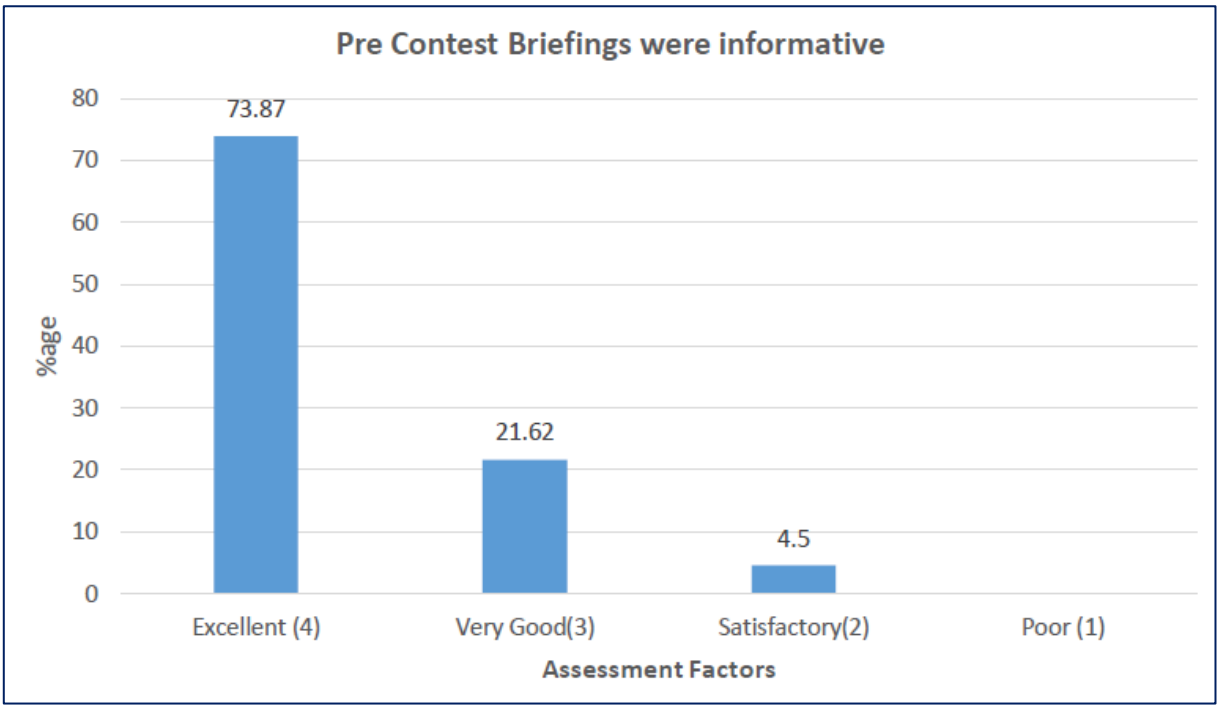
72.22% of the speakers found that contests were Excellent in terms of usefulness, 22.22% felt that it was very good, while 5.56% found it satisfactory, according to the feedback analysis.

14.3. Visitors Feedback Analysis- Overall Analysis



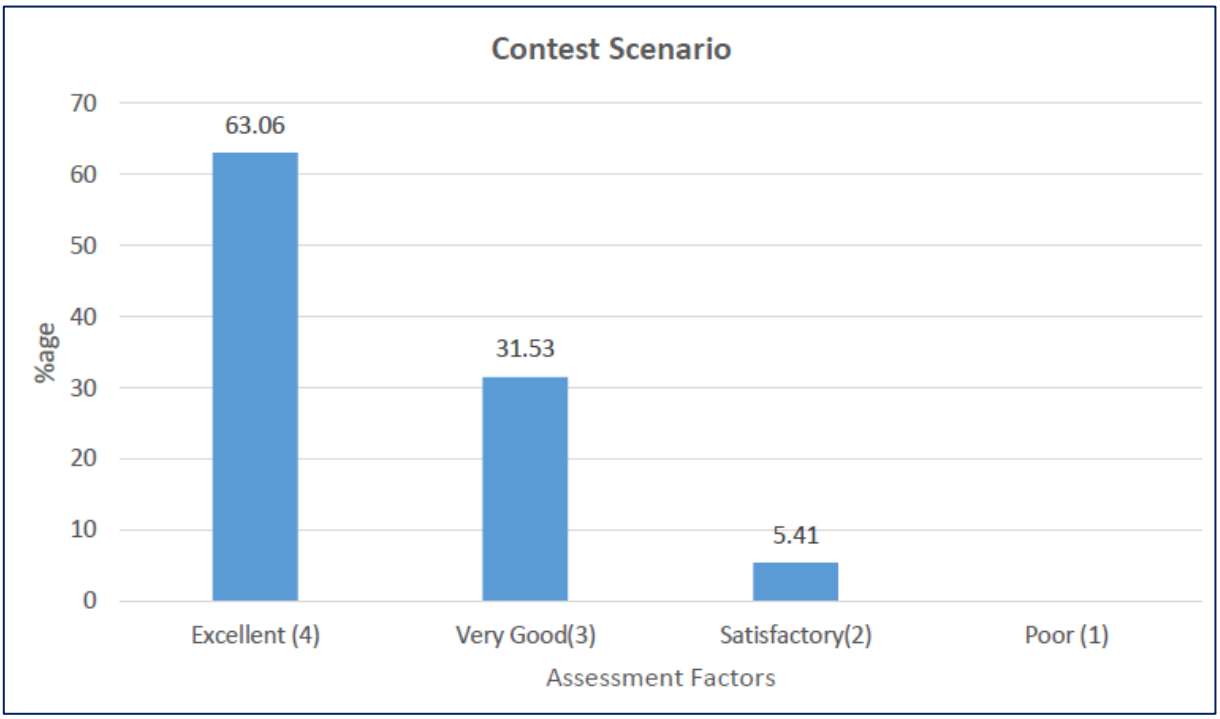
Analysing the overall feedbacks given by the visitors we found that on average 64.21% of the visitors found the entire event perfectly organised and rated mostly assessments as excellent. 27.15% visitors on average rated the event very good, 8.01% visitors on average rated the organized festival as satisfactory and rest 0.90% on average found the usefulness of the event unsatisfactory.

14.3.1. Pre-Contest Briefings were informative and provided the necessary information



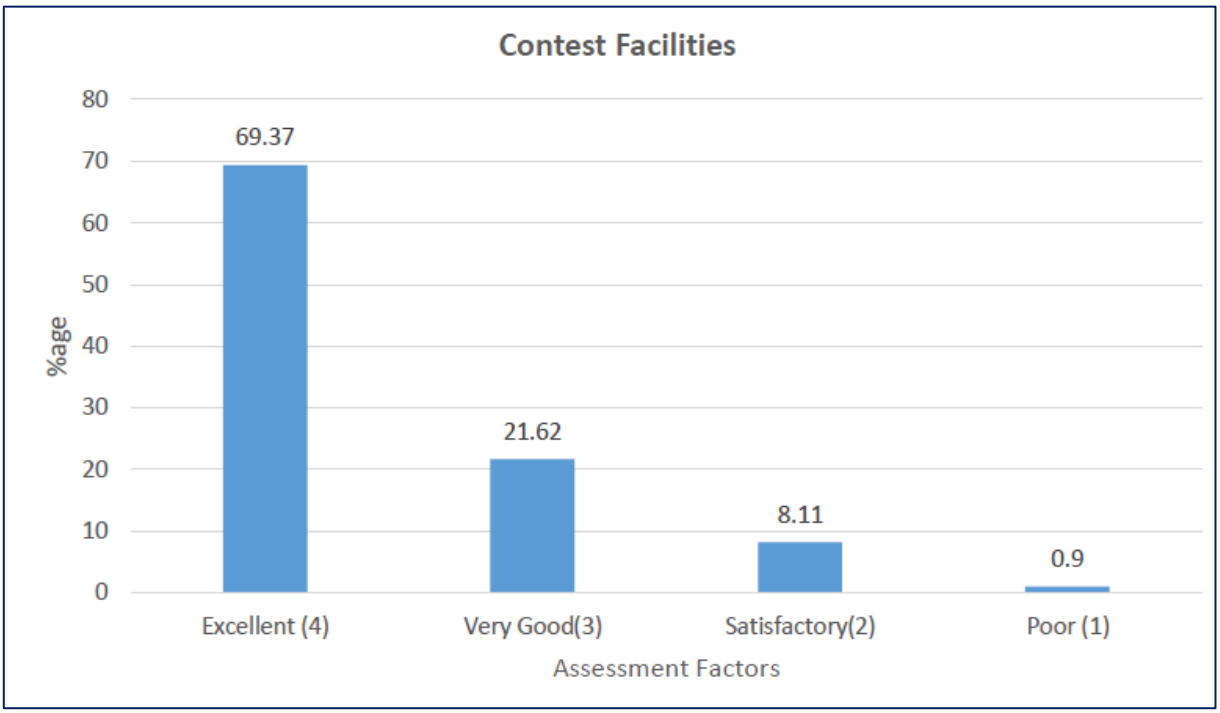
The Visitors had shared their inputs for the pre-contest briefings, as illustrated in graph 73.87% participants had an opinion that the event was excellently briefed. 21.62 % of participants felt that briefing of the festival was very good. On the other hand 4.5% participants felt that the briefing was satisfactory.

14.3.2. Contest Scenario



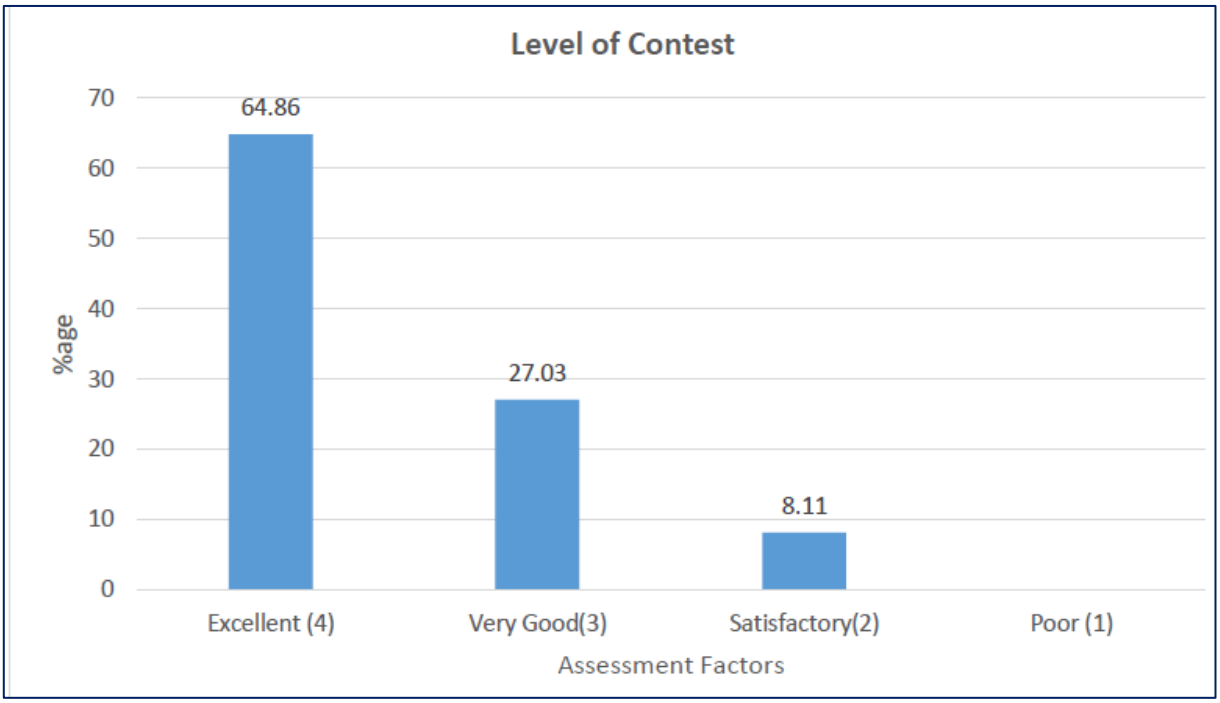
As per the feedback of the visitors, the above graph shows that 63.06% of them gave excellent reviews for the contests scenario and 31.53% of them felt that contests were very good while 5.41% of visitors found it satisfactory.

14.3.3. Contest Facilities



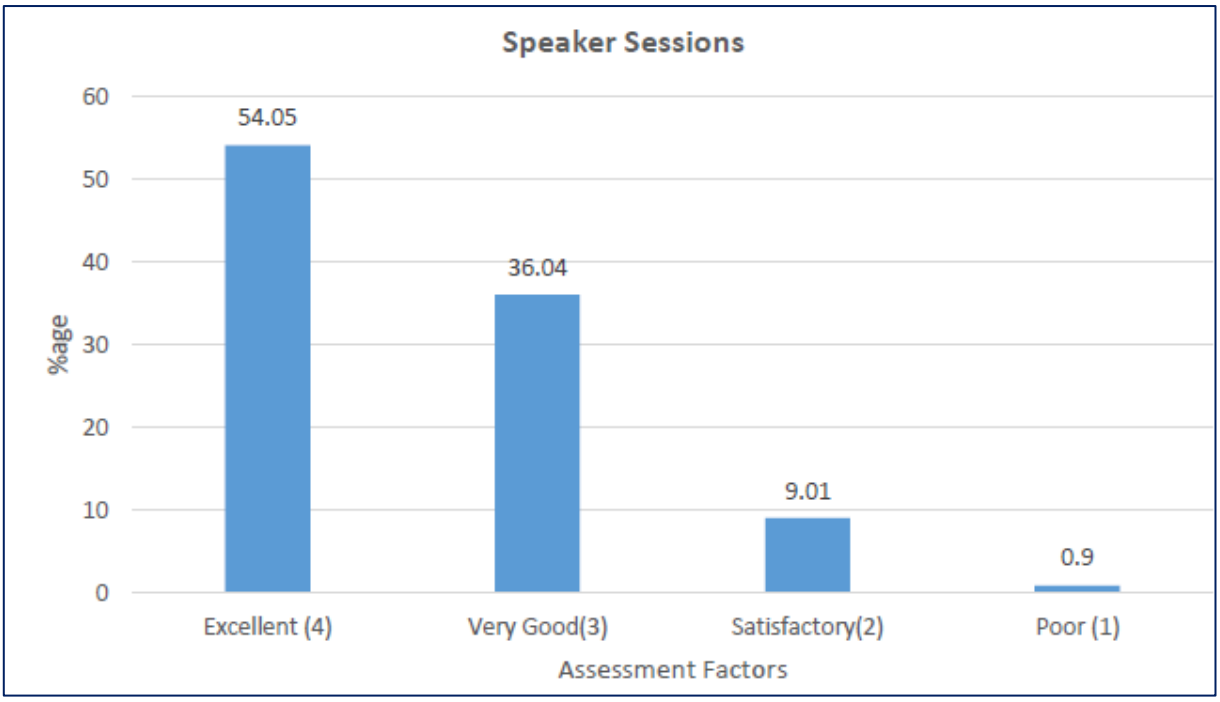
According to 69.37% of the visitors, contest facilities were excellent, while according to 21.67 % of the visitors facilities provided were very good and 8.11% of them felt it was satisfactory while 0.9% of the visitors found the facilities were poor.

14.3.4. Level of Contest



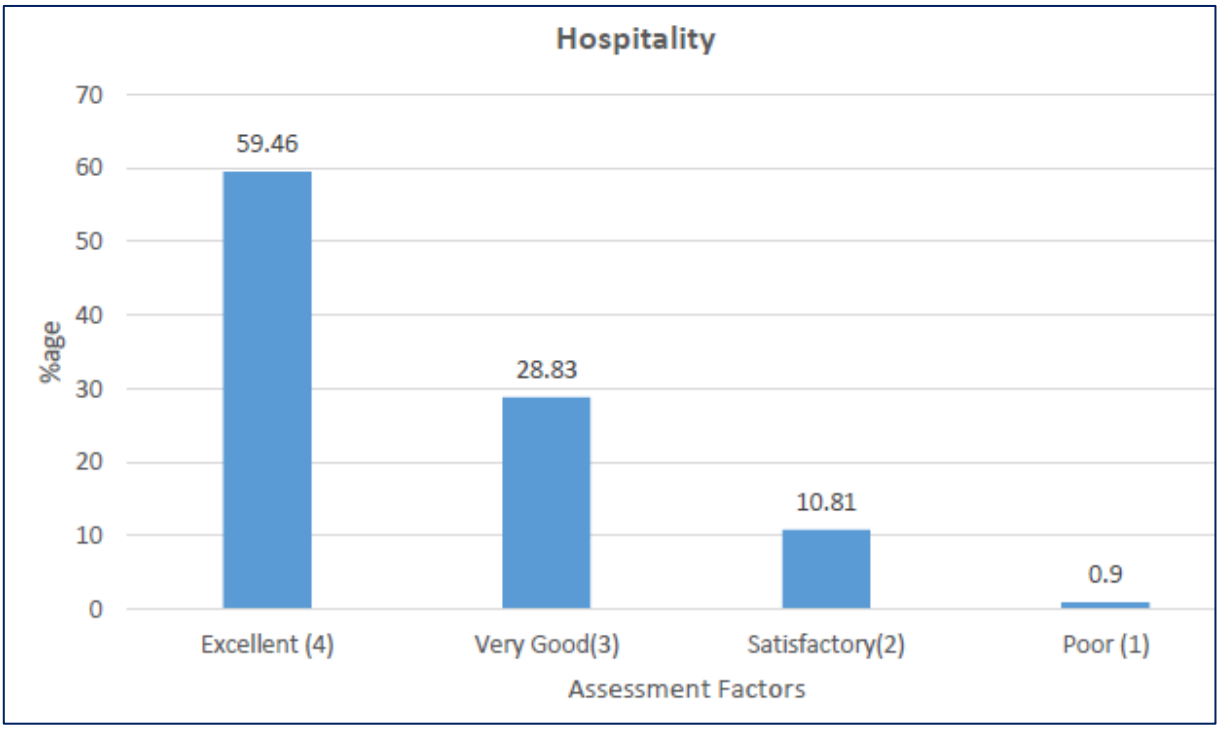
As per the feedback, 64.86% of visitors gave excellent ratings for the level of contests, 27.03% of them found the level of contests were very good, while 8.11% of the visitors level of contests were satisfactory.

14.3.5. Speaker Sessions



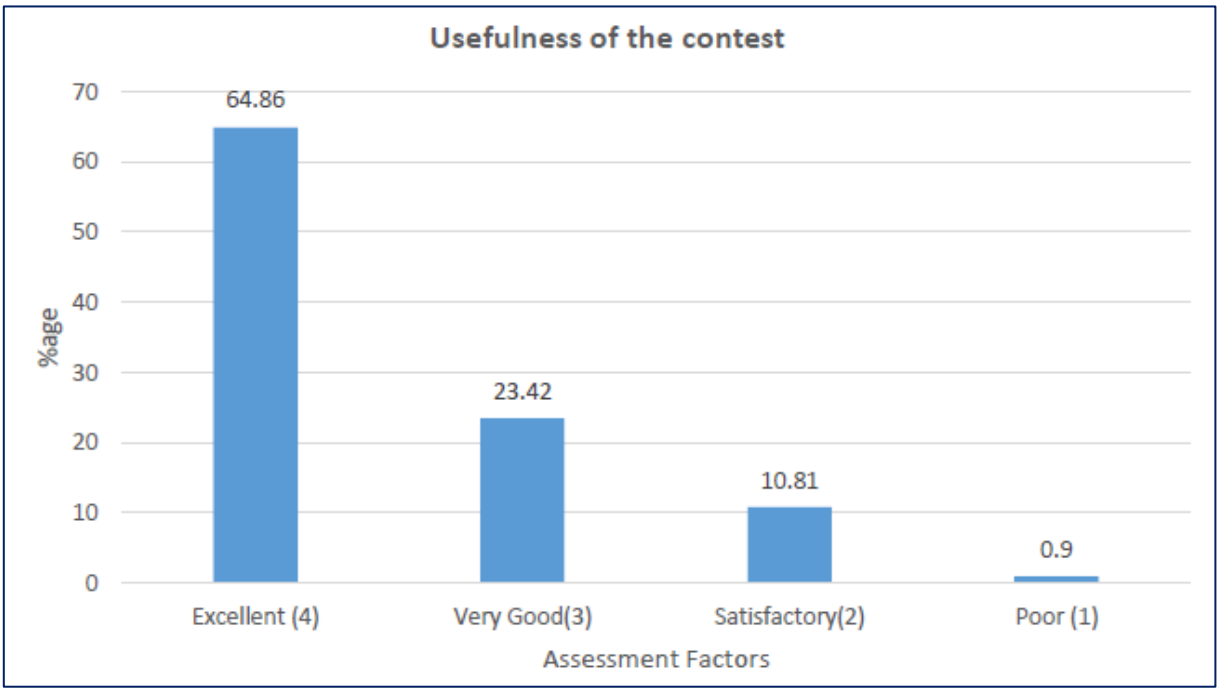
As per the above graph we can extract the details that 54.05% felt that the speakers' sessions were excellent. 36.04% felt that it was good and 9.01% found the sessions satisfactory whereas 0.9% found the sessions poorly conducted.

14.3.6. Hospitality



As mentioned in the above graph, hospitality provided in the festival was excellent according to the 59.46% of the visitors, very good according to the 28.83% of the visitors while 10.81 % of them felt that the hospitality provided was excellent and 0.9% of them found it poor.

14.3.7. Usefulness of the contest



In the opinion of 64.86% of the visitors the was very much useful and they gave Excellent rating, 23.42% found it Very good ,while 10.81% felt satisfactory and 0.9% of the visitors felt it was not as much useful and gave poor rating.

15. KEY CONSIDERATIONS

It is indeed a great honour for BDO India to be associated with esteemed organization like DARC - ITDA as a knowledge partner for such a prestigious event INDIA DRONE FESTIVAL 2.0 – DRONATHON@DRONANAGARI 2020.

We saw a very good participation from industry, government departments, and most importantly from student community – the future of the country. During the event we had one-on-one interactions with industry participant and tried to understand their outlook towards Drone Industry as a whole, potential area of growth both in terms of producer and market, and area of concerns related to Drone industry growth. We also tried to understand their general expectations from government of the state, and enablers that will benefit both the industry and citizens. We also had discussions with our BDO global counterparts, and experts in drone area to understand the area of concerns from industry and possible enablers that government can provide.

Here are some key considerations which may help Drone industry prosper in Uttarakhand to make Uttarakhand as a leader in Drone Manufacturing and development hub –

- Given the geographical terrain, Uttarakhand may have multiple use cases for providing efficient services using drone technology. A favourable industrial policy will help the industry to grow in state.
- During the interactions with industry experts, they expressed their concerns about multiple permissions for implementation and usage of drone in the State. One major issue is also related to sites for testing drones related activities. It will be certainly helpful to have streamlined processes and single window unified platform for drone industry which may facilitate all stakeholders including Government, industry and citizens.
- DARC- ITDA may also setup drone information and training centers in collaboration with CSCs for training the youth of State. This will create a future ready workforce in State which may be used for providing services within and outside State.
- Uttarakhand is a tourist hub and tourism sector is contributing significantly in revenue generation for the State. Drone can be used by the stakeholders of this sector like tourism operators, hospitality industry, photography, videography of tourist sites by tourists themselves with help from local drone operators.
- Special provisions and incentive may be considered for the start-ups and MSMEs who want to excel in this industry. ITDA can enable these agencies with help from DARC for providing technology, training and industry outreach.
- DARC- ITDA also need to create end to end ecosystem for Drone and its allied industries with starting from manufacturing, policies, implementation, support and capacity building
- DARC may also join hands with leading technology institutes across country and globally for technology upgradation which may help them to create innovative solutions.
- State government may consider creating a special policy including allocation of land and tax exemptions of organizations working in drone industry.
- DARC may form an SPV in collaboration with industry on PPP model wherein new product development may be initiated for providing professional services to government, semi government and industries.
- Drones can be effectively used in disaster management, agriculture/ horticulture production, deforestation control and many other sectors. DARC may also device sectoral SOPs for usage of drones in respective sector.
- DARC may introduce drone manufacturing lab for usage of students of State. This may also act as an alternative for students in school workshops.